TEXTILE ORDERS PERFORMANCE DASHBOARD | JAN - SEP 2026

MONTH	FACTORY	CATEGORY	REGION	SALESPERSON
All	All	All	All	All

Reset: set every drop-down to All

REVENUE (USD)
$20,124,550
100.0% of Jan-Sep total | 995 orders

GROSS MARGIN %
56.2%
Target 57.4% | margin $11,316,708

TARGET ATTAINMENT
95.3%
-$985,696 vs target $21,110,246

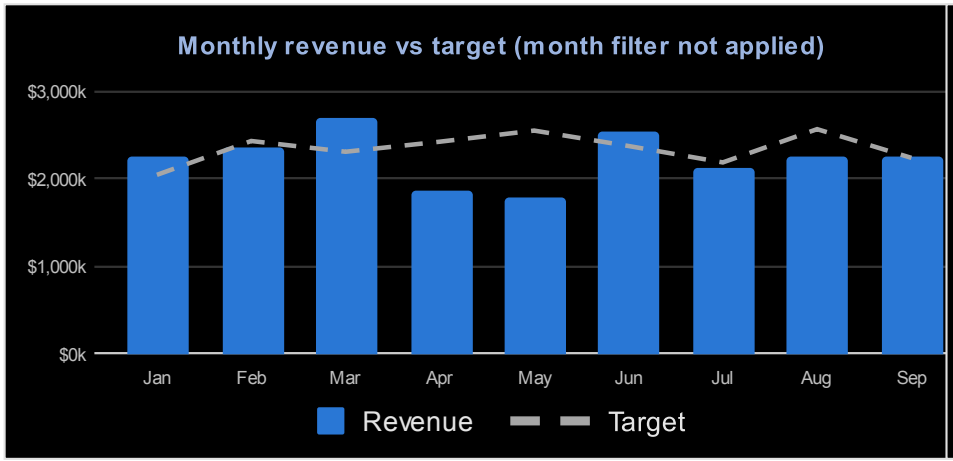
ORDER LINES
995
2,590,605 units | avg order $20,226

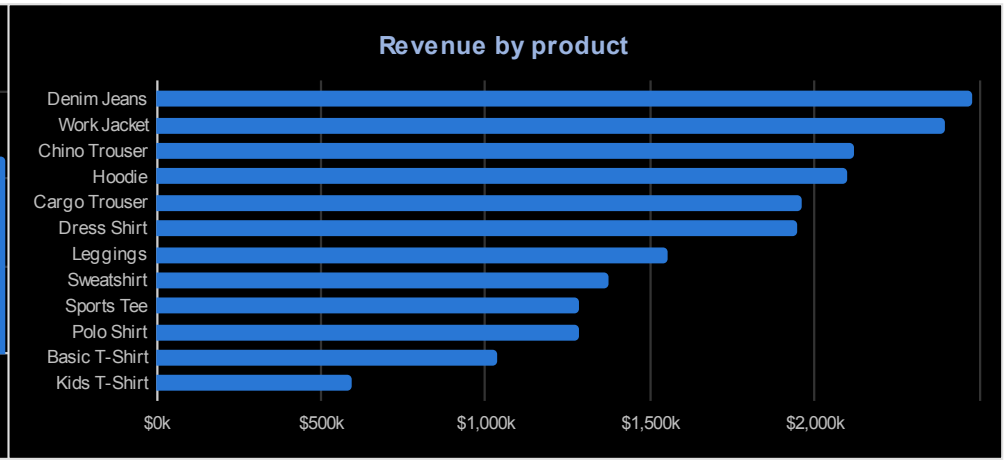
ON-TIME DELIVERY
31.2%
685 late orders | $13,875,460 shipped late

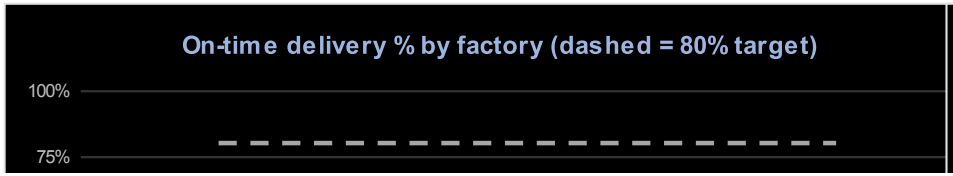
DEFECT RATE (UNITS)
2.50%
64,756 defective units

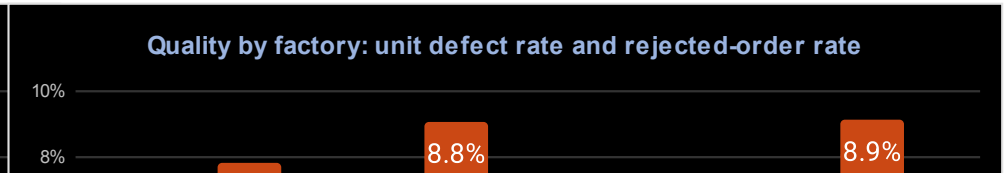
REJECTED ORDERS
7.7%
77 orders | $1,583,938 revenue at risk

AVG DAYS LATE (LATE ORDERS)
9.6 days
Fill rate 97.2% | lead time 51 days





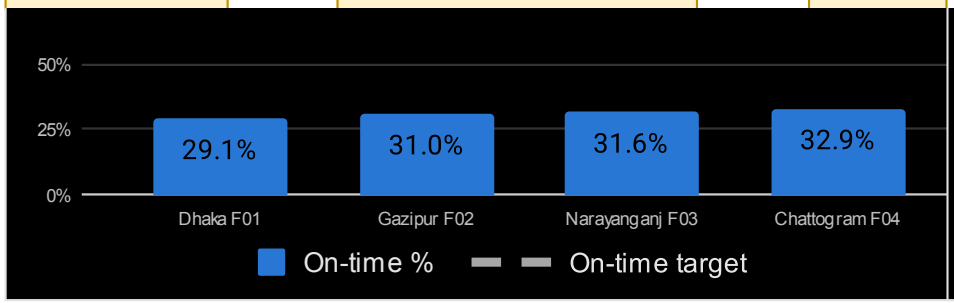


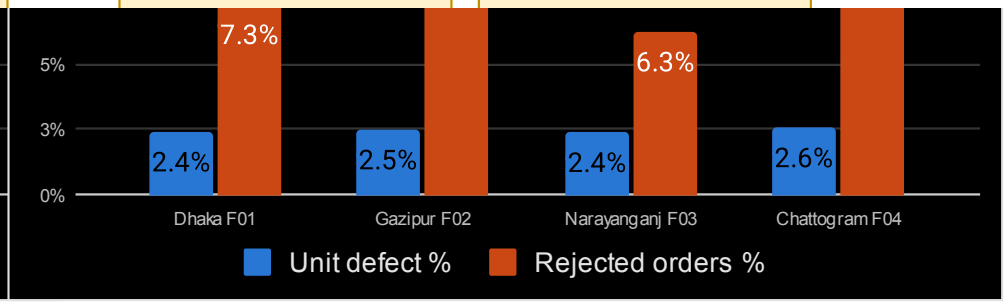


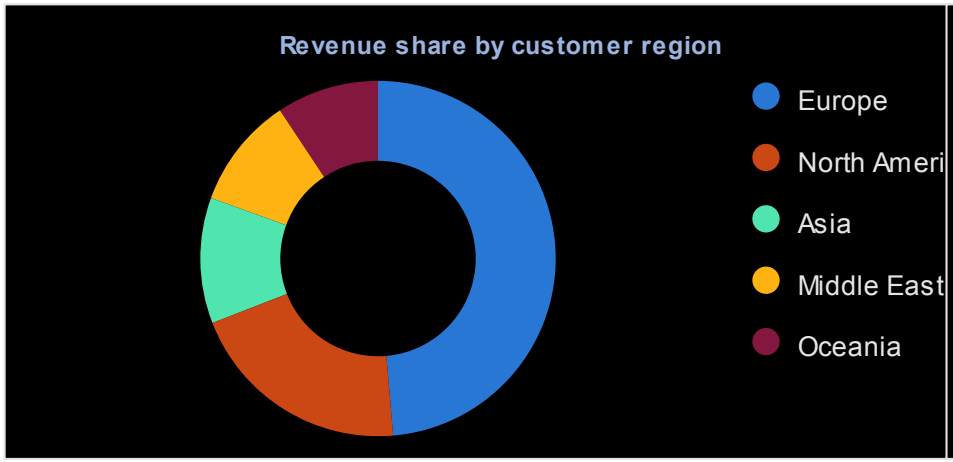
TEXTILE ORDERS PERFORMANCE DASHBOARD | JAN - SEP 2026

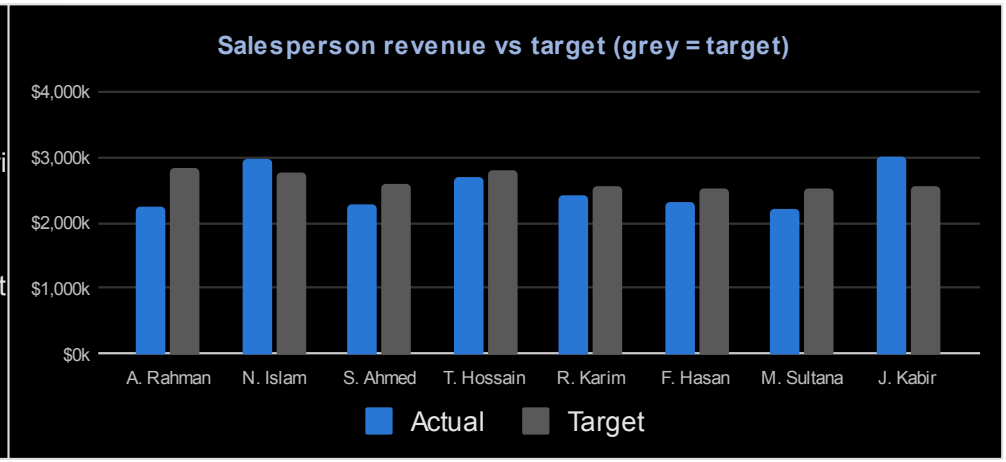
MONTH: All | FACTORY: All | CATEGORY: All | REGION: All | SALESPERSON: All

Reset: set every drop-down to All









Top 5 products

#	Product	Revenue	Share of filtered revenue
1	Denim Jeans	$2,480,669	12.3%
2	Work Jacket	$2,394,298	11.9%

Delivery and quality mix

Delivery bucket	Orders	Share
On time / early	310	31.2%
1-7 days late	261	26.2%

Quality	Orders	Share
Passed	761	76.5%
Rework	157	15.8%

TEXTILE ORDERS PERFORMANCE DASHBOARD | JAN - SEP 2026

MONTH	FACTORY	CATEGORY	REGION	SALESPERSON
All	All	All	All	All

Reset: set every drop-down to All

3	Chino Trouser	$2,116,932	10.5%
4	Hoodie	$2,100,395	10.4%
5	Cargo Trouser	$1,958,031	9.7%

8-14 days late	279	28.0%
15+ days late	145	14.6%

Rejected	77	7.7%

Equivalent PivotTable build: PivotCharts on Clean_Orders + slicers (Month, Factory, Category, Region, Salesperson) connected to every pivot via Report Connections, plus a Timeline on Order_Date.

Task 6 - Data-Driven Insights & Recommendations

#	Theme	Type	Finding (evidence)	Why it matters	Recommendation	Priority	Evidence
1	**Delivery reliability**	Problem	Only 31.2% of 995 order lines shipped on or before the promised date. 685 late orders carry $13,875,460 (69%) of revenue; late orders average 9.6 days late and 424 (43%) are more than a week late.	Chronic lateness invites buyer penalties, discounts and air-freight costs, and puts repeat programmes at risk. It is the largest performance gap in the data.	Make on-time delivery the headline operations KPI: weekly review by factory and buyer, with early escalation of any order forecast to miss its promise date.	**High**	Ops_Analysis
2	**Root cause of lateness**	Problem	Customers are promised 45.1 days on average, but real order-to-ship lead time is 50.6 days (+5.5 days). Every factory sits between 29% and 33% on-time, so lateness is built in when dates are quoted - a planning issue, not one weak plant.	Because the gap is systemic, pushing one factory harder will not fix it; the promise itself is unrealistic.	Quote realistic dates: adding 13 buffer days (a promise window of about 58 days) lifts on-time delivery to 81%. Alternatively cut real lead time by the same number of days through capacity and material planning (Advanced_Analysis F).	**High**	Advanced_Analysis
3	**Quality losses**	Problem	Unit defect rate is 2.50% (64,756 units, about $220,165 of cost). 77 orders (7.7%) were rejected, putting $1,583,938 of revenue at risk, and 157 more (15.8%) needed rework. Rejected lots average 6.0% defects against 2.2% for passed lots.	Rejections trigger chargebacks, re-shipment and lost margin. A lot at roughly 6% defects is effectively a rejected lot.	Introduce inline inspection at sewing and finishing with a stop-and-fix rule well below 6% defects, plus a pre-final AQL audit on high-risk styles.	**High**	Ops_Analysis
4	**Factory and product hotspots**	Problem	Chattogram Finishing Unit has the highest rejected-order rate (8.9%) and also the highest in-process defect rate (3.24%). Narayanganj Woven Unit is best on rejections (6.3%). Finishing is the weakest process (3.21% defects) and Denim Jeans the most-rejected product (11.0% of its orders).	Quality problems are concentrated, so a targeted project at the weakest unit and process gives the fastest payback.	Run a root-cause (Pareto / fishbone) project at the weakest unit, starting with its worst process and the most-rejected styles; transfer practices from the best-performing unit.	**High**	Ops_Analysis
5	**Margin leakage**	Problem	Every one of the 995 orders is below its product target margin: actual 56.2% vs a revenue-weighted target of 57.4% (-1.13 pts, about $227,507). Prices are already 8.2% above standard, so the shortfall is cost-driven. Work Jacket has the thinnest margin (51.8%).	A systematic shortfall on every order points to costing assumptions (fabric, CM, rework) rather than a few bad deals.	Re-cost bills of material and CM for all styles. Goal Seek shows a -2.6% unit-cost change, or a +2.7% price change, restores target margin (Advanced_Analysis D).	**High**	Advanced_Analysis
6	**Revenue vs target**	Trend	Jan-Sep revenue of $20,124,550 is 95.3% of the $21,110,246 target (-$985,696). The shortfall is concentrated in May 2026 (70% of target) and Apr 2026 (77%); Mar 2026 was strongest (117%). Only 37 of 72 salesperson-months met target.	Strong months cannot offset the spring trough; targets also swing sharply month to month, so attainment is volatile.	Build pipeline for the April-May trough (early-booking incentives for autumn/winter programmes) and review the monthly phasing of targets with the sales team.	**High**	Sales_Analysis
7	**Salesperson performance**	Pattern	J. Kabir (118% of target) and N. Islam (107%) lead on attainment, while A. Rahman trails at 79% (-$586,195).	Results vary widely across a team selling the same products from the same factories, so account management practice matters.	Pair the lowest-attainment salespeople with the top performers on key accounts; agree account plans for the weak months.	**Medium**	Sales_Analysis
8	**Revenue mix**	Opportunity	Woven garments earn 54% of revenue from 44% of orders (average $9.45 per unit vs $6.42 for knit). The top 4 products deliver 45% of revenue and Europe 49%. Customer concentration is low: the largest customer (Sakura Lifestyle) is 6.5% of revenue and the top 5 together 31%.	Higher-value woven programmes are the growth lever, and the diversified customer base limits dependency risk.	Grow woven programmes (denim, jackets, chinos) where capacity allows, but fix their rejection rates first; keep the broad customer base.	**Medium**	Sales_Analysis

Task 6 - Data-Driven Insights & Recommendations

#	Theme	Type	Finding (evidence)	Why it matters	Recommendation	Priority	Evidence
9	**Short shipments and production**	Problem	170 orders were shipped short: fill rate is 97.2%, leaving 73,580 units ($578,631) undelivered. Production runs below plan too: 98.2% plan attainment, 62% of batches under plan, and rework recovers only 27% of defects.	Short shipments are lost revenue and a common cause of buyer claims; they compound the lateness problem.	Plan a 2-3% production allowance on cut quantities and track OTIF (on time, in full) rather than on-time alone.	**Medium**	Ops_Analysis
10	**Data quality**	Process	30 data-quality issues were found and fixed (duplicates, placeholders such as 9,999,999 and 999, text dates, unmatched IDs, inconsistent labels). Left uncorrected, revenue would have been overstated by $10,117,700 (50%).	Management decisions were at risk of being made on numbers inflated by about half.	Add input controls at source: drop-down lists for IDs and labels, date validation, and a monthly duplicate/ outlier check before reporting.	**Medium**	Cleaning_Log

Task 7 - Analytical Report: Textile Orders, Quality & Delivery (Jan-Sep 2026)

1. Problem and data overview

The business needs to understand commercial results, production quality and delivery performance for its apparel export orders from January to September 2026, and to decide where to act first. Raw_Orders holds 1,001 rows across 31 fields, supported by master tables (12 products, 20 customers, 4 factories, 8 salespeople), 72 salesperson-month revenue targets and 300 production batches. After cleaning, 995 valid order lines remain, worth $20,124,550 in revenue and 2,590,605 units.

Questions answered: (1) Are revenue and margin on target? (2) Which products, customers, markets, factories and salespeople drive or drag performance? (3) How good are quality and on-time delivery, and why? (4) What should management do next?

2. Methodology

- Cleaning: the raw sheet was left untouched and a cleaned copy (Clean_Orders) built, with every change logged row by row in Issue_Register - 30 issues: 1 non-data row, 5 duplicate rows, 7 missing values, 5 invalid/outlier values, 5 inconsistent labels/IDs, 4 unmatched IDs and 3 text dates. Values were recovered from arithmetic identities (Revenue = Qty x Price; Lead time = Ship - Order) or master tables wherever they were fully determined; otherwise the field was left blank and excluded only from the affected metric. Bridges in Cleaning_Log reconcile raw to clean rows, revenue, units and margin.

- Formulas: 30 helper columns in Clean_Orders (INDEX/MATCH lookups to the masters, revenue check, margin % and gap to target, defect rate, days late, on-time flag, status and lead-time validation, fill rate, month key, delivery buckets). Named ranges keep every summary formula readable and auditable.

- Analysis: SUMIFS/COUNTIFS summary tables, equivalent to PivotTables, by product, category, customer, region, country, salesperson, month, channel, payment terms, season and factory. Production_Data was related through Factory_ID and Product_ID for a combined factory scorecard, and targets were joined on Salesperson_ID + month.

- Advanced analysis and dashboard: an input-driven what-if model with five scenarios and a selector, Goal Seek answers, two-way sensitivity tables and a delivery-buffer simulation; an interactive dashboard where five drop-down filters drive eight KPI cards, six charts and two tables.

3. Key findings

- Only 31.2% of 995 order lines shipped on or before the promised date. 685 late orders carry $13,875,460 (69%) of revenue; late orders average 9.6 days late and 424 (43%) are more than a week late.

- Customers are promised 45.1 days on average, but real order-to-ship lead time is 50.6 days (+5.5 days). Every factory sits between 29% and 33% on-time, so lateness is built in when dates are quoted - a planning issue, not one weak plant.

- Unit defect rate is 2.50% (64,756 units, about $220,165 of cost). 77 orders (7.7%) were rejected, putting $1,583,938 of revenue at risk, and 157 more (15.8%) needed rework. Rejected lots average 6.0% defects against 2.2% for passed lots.

- Every one of the 995 orders is below its product target margin: actual 56.2% vs a revenue-weighted target of 57.4% (-1.13 pts, about $227,507). Prices are already 8.2% above standard, so the shortfall is cost-driven. Work Jacket has the thinnest margin (51.8%).

- Jan-Sep revenue of $20,124,550 is 95.3% of the $21,110,246 target (-$985,696). The shortfall is concentrated in May 2026 (70% of target) and Apr 2026 (77%); Mar 2026 was strongest (117%). Only 37 of 72 salesperson-months met target.

- Woven garments earn 54% of revenue from 44% of orders (average $9.45 per unit vs $6.42 for knit). The top 4 products deliver 45% of revenue and Europe 49%. Customer concentration is low: the largest customer (Sakura Lifestyle) is 6.5% of revenue and the top 5 together 31%.

4. Insights

- Delivery, not demand, is the binding constraint. Promise windows (45 days) are shorter than real lead times (51 days) at every factory, so most orders are late from the moment they are booked.

- Quality losses are concentrated and therefore fixable: rejected lots cluster at about 6% defects, and the worst factory, process and styles are clearly identified.

- Margin is leaking through cost, not price: prices run 8.2% above standard, yet the portfolio earns 1.1 pts less than target margin.

- Sales momentum dipped in April-May; most other months were close to or above target, so the year-to-date gap is recoverable.

- Revenue is well spread across customers (low concentration risk) but tilted towards woven products and Europe, which sets priorities for capacity and quality investment.

5. Recommendations (in priority order)

1. Quote realistic dates: adding 13 buffer days (a promise window of about 58 days) lifts on-time delivery to 81%. Alternatively cut real lead time by the same number of days through capacity and material planning (Advanced_Analysis F).

2. Make on-time delivery the headline operations KPI: weekly review by factory and buyer, with early escalation of any order forecast to miss its promise date.

3. Run a root-cause (Pareto / fishbone) project at the weakest unit, starting with its worst process and the most-rejected styles; transfer practices from the best-performing unit.

4. Re-cost bills of material and CM for all styles. Goal Seek shows a -2.6% unit-cost change, or a +2.7% price change, restores target margin (Advanced_Analysis D).

5. Build pipeline for the April-May trough (early-booking incentives for autumn/winter programmes) and review the monthly phasing of targets with the sales team.

6. Plan a 2-3% production allowance on cut quantities and track OTIF (on time, in full) rather than on-time alone.

6. Conclusion

The business is close to its revenue plan (95.3% of target) and keeps a healthy 56.2% gross margin, but operational reliability is weak: 69% of orders ship late and 7.7% are rejected. Realistic promise dates and a focused quality programme offer the fastest gains. The Recovery-plan scenario (+3% price, +2% volume, -1% unit cost, 25% fewer defects) lifts revenue to $21,142,852 (100% of target) and gross margin by $987,566.

Task 1 - Data Quality Issue Register (row-level)

#	Raw row	Order_ID	Field	Issue category	Issue detail	Original value	Cleaned value	Treatment applied	Justification	Alternative treatment	How to detect in Excel
1	1002	(none)	Entire row	Irrelevant record	Irrelevant / non-data record	Text notice in Order_ID column; all other fields blank	Row removed	Delete row	Not an order line: no ID, date, customer or values. Would inflate counts and break date/number columns.	-	Filter Order_ID: text that is not an ORD##### ID; all other fields blank (COUNTBLANK on the row = 30).
2	44	ORD10042	Entire row	Duplicate record	Duplicate record	Exact copy of raw row 43 (all 31 fields identical)	Row removed	Keep first occurrence, delete copy	Order_ID is the unique order-line key and every field matches, so the copy double-counts revenue, quantity and margin.	Excel Data > Remove Duplicates on all columns gives the same result.	Helper =COUNTIF($A$2:A2,A2)>1 flags repeat IDs; compare all fields (COUNTIFS) or Data > Remove Duplicates on all columns.
3	179	ORD10177	Entire row	Duplicate record	Duplicate record	Exact copy of raw row 178 (all 31 fields identical)	Row removed	Keep first occurrence, delete copy	Order_ID is the unique order-line key and every field matches, so the copy double-counts revenue, quantity and margin.	Excel Data > Remove Duplicates on all columns gives the same result.	Helper =COUNTIF($A$2:A2,A2)>1 flags repeat IDs; compare all fields (COUNTIFS) or Data > Remove Duplicates on all columns.
4	390	ORD10388	Entire row	Duplicate record	Duplicate record	Exact copy of raw row 389 (all 31 fields identical)	Row removed	Keep first occurrence, delete copy	Order_ID is the unique order-line key and every field matches, so the copy double-counts revenue, quantity and margin.	Excel Data > Remove Duplicates on all columns gives the same result.	Helper =COUNTIF($A$2:A2,A2)>1 flags repeat IDs; compare all fields (COUNTIFS) or Data > Remove Duplicates on all columns.
5	723	ORD10721	Entire row	Duplicate record	Duplicate record	Exact copy of raw row 722 (all 31 fields identical)	Row removed	Keep first occurrence, delete copy	Order_ID is the unique order-line key and every field matches, so the copy double-counts revenue, quantity and margin.	Excel Data > Remove Duplicates on all columns gives the same result.	Helper =COUNTIF($A$2:A2,A2)>1 flags repeat IDs; compare all fields (COUNTIFS) or Data > Remove Duplicates on all columns.
6	904	ORD10902	Entire row	Duplicate record	Duplicate record	Exact copy of raw row 903 (all 31 fields identical)	Row removed	Keep first occurrence, delete copy	Order_ID is the unique order-line key and every field matches, so the copy double-counts revenue, quantity and margin.	Excel Data > Remove Duplicates on all columns gives the same result.	Helper =COUNTIF($A$2:A2,A2)>1 flags repeat IDs; compare all fields (COUNTIFS) or Data > Remove Duplicates on all columns.
7	69	ORD10068	Customer_Name	Missing value	Missing value	(blank)	Urban Loom Retail	Fill from Customer_Master via Customer_ID (INDEX/MATCH or XLOOKUP)	Customer_ID is valid, so the name is fully determined by the master table.	-	Filter (Blanks) on each column, =COUNTBLANK(range), or Go To Special > Blanks.
8	123	ORD10122	Salesperson_ID	Missing value	Missing value	(blank)	S03	Fill from Salesperson_Master using the Salesperson name on the row	Name is present and matches master exactly (name-ID pairs are consistent on all other rows).	Flag as "Unassigned" (revenue excluded from salesperson views).	Filter (Blanks) on each column, =COUNTBLANK(range), or Go To Special > Blanks.
9	216	ORD10215	Unit_Price_USD	Missing value	Missing value	(blank)	8.77	Impute as Revenue_USD / Order_Qty (33,554.02 / 3,826)	Revenue and quantity are present and the division gives an exact 2-decimal price within the product price band.	-	Filter (Blanks) on each column, =COUNTBLANK(range), or Go To Special > Blanks.
10	318	ORD10317	Defect_Qty	Missing value	Missing value	(blank)	(blank)	Leave blank; exclude this order from defect-rate denominators	Defect count cannot be derived from other fields; imputing 0 would understate defects.	Impute the product median or 0 (impact < 0.01 pp on defect rate).	Filter (Blanks) on each column, =COUNTBLANK(range), or Go To Special > Blanks.
11	505	ORD10504	Actual_Ship_Date	Missing value	Missing value	(blank)	2026-02-20	Impute as Order_Date + Lead_Time_Days (2026-01-02 + 49)	Lead_Time_Days = Ship - Order on every complete row, so the ship date is recoverable. Result (16 days after promise) agrees with the recorded status 'Late'.	Leave blank and exclude from on-time % (on-time % then uses 994 orders).	Filter (Blanks) on each column, =COUNTBLANK(range), or Go To Special > Blanks.
12	646	ORD10645	Lead_Time_Days	Missing value	Missing value	(blank)	69	Recalculate as Actual_Ship_Date - Order_Date	Field is fully determined by the two dates (identity holds on all complete rows).	-	Filter (Blanks) on each column, =COUNTBLANK(range), or Go To Special > Blanks.
13	813	ORD10812	Season	Missing value	Missing value	(blank)	Unknown	Label as 'Unknown' (keep the row)	Season is a buyer-season label that does not follow order month, so it cannot be inferred; row is valid for all other metrics.	Leave blank, or exclude the row from season views only.	Filter (Blanks) on each column, =COUNTBLANK(range), or Go To Special > Blanks.
14	335	ORD10334	Order_Qty	Invalid / anomalous value	Invalid value (negative quantity)	-250	4397	Recover as Revenue_USD / Unit_Price_USD (53,995.16 / 12.28)	Quantity cannot be negative. Revenue / price gives exactly 4,397, which also equals Delivered_Qty, so the true quantity is recoverable. Also clears the Defect_Qty > Order_Qty violation.	Exclude the row instead (removes $53,995 revenue).	Rule flag =IF(OR(Qty<=0,Price<=0,Defect>Qty),"Check","OK"); sort descending (spots 9,999,999 and 999); Revenue vs Qty x Price check.
15	446	ORD10445	Unit_Price_USD	Invalid / anomalous value	Invalid value (zero price)	0	10.44	Impute as Revenue_USD / Order_Qty (8,811.36 / 844)	A zero price contradicts recorded revenue and margin; the division gives an exact price within the product band.	-	Rule flag =IF(OR(Qty<=0,Price<=0,Defect>Qty),"Check","OK"); sort descending (spots 9,999,999 and 999); Revenue vs Qty x Price check.
16	557	ORD10556	Revenue_USD	Invalid / anomalous value	Outlier / placeholder value	9999999	14613.7	Recalculate as Order_Qty x Unit_Price_USD (1,585 x 9.22)	9,999,999 is a placeholder (next-largest revenue is ~$57k) and would inflate total revenue by ~$10M; quantity, price and margin are all normal.	Exclude the row instead.	Rule flag =IF(OR(Qty<=0,Price<=0,Defect>Qty),"Check","OK"); sort descending (spots 9,999,999 and 999); Revenue vs Qty x Price check.
17	779	ORD10778	Delivered_Qty	Invalid / anomalous value	Invalid value (zero delivered)	0	(blank)	Treat as unknown (blank); exclude from fill-rate denominators	Order shipped on 2026-04-26 with status 'On Time', so 0 delivered is impossible; true quantity unknown.	Assume full delivery (= Order_Qty).	Rule flag =IF(OR(Qty<=0,Price<=0,Defect>Qty),"Check","OK"); sort descending (spots 9,999,999 and 999); Revenue vs Qty x Price check.

Task 1 - Data Quality Issue Register (row-level)

#	Raw row	Order_ID	Field	Issue category	Issue detail	Original value	Cleaned value	Treatment applied	Justification	Alternative treatment	How to detect in Excel
18	890	ORD10889	Lead_Time_Days	Invalid / anomalous value	Invalid / outlier value	999	65	Recalculate as Actual_Ship_Date - Order_Date	999 is a placeholder (next-largest value is 82 days); the dates give the true lead time.	Exclude from lead-time averages.	Rule flag =IF(OR(Qty<=0,Price<=0,Defect>Qty),"Check","OK"); sort descending (spots 9,999,999 and 999); Revenue vs Qty x Price check.
19	93	ORD10092	Channel	Inconsistent label / format	Inconsistent label	Direct buyer	Direct Buyer	Standardise case (PROPER/Find & Replace)	Same category written in different case; would split pivot rows.	-	Filter drop-down or =UNIQUE(range) lists the variants; EXACT/PROPER/TRIM; Find & Replace.
20	157	ORD10156	Quality_Status	Inconsistent label / format	Inconsistent label	passed	Passed	Standardise case (PROPER/Find & Replace)	Same category written in different case; would split pivot rows.	-	Filter drop-down or =UNIQUE(range) lists the variants; EXACT/PROPER/TRIM; Find & Replace.
21	268	ORD10267	Salesperson_ID	Inconsistent label / format	Inconsistent ID format + conflicting name	s-03	S05	Re-key from the Salesperson name (R. Karim = S05)	's-03' is a malformed ID (lower case + hyphen). Re-formatting gives S03 (S. Ahmed) but the row's name is R. Karim (S05). Name-ID pairs agree on every other row, so the name is treated as the reliable field.	S03 by format alone, although that contradicts the R. Karim name on the row.	Filter drop-down or =UNIQUE(range) lists the variants; EXACT/PROPER/TRIM; Find & Replace.
22	479	ORD10478	Channel	Inconsistent label / format	Inconsistent label	Buying house	Buying House	Standardise case (PROPER/Find & Replace)	Same category written in different case; would split pivot rows.	-	Filter drop-down or =UNIQUE(range) lists the variants; EXACT/PROPER/TRIM; Find & Replace.
23	701	ORD10700	Quality_Status	Inconsistent label / format	Inconsistent label	REWORK	Rework	Standardise case (PROPER/Find & Replace)	Same category written in different case; would split pivot rows.	-	Filter drop-down or =UNIQUE(range) lists the variants; EXACT/PROPER/TRIM; Find & Replace.
24	614	ORD10613	Customer_ID	Unmatched ID	Unmatched ID	CUS999	CUS012	Re-key from Customer_Name via Customer_Master	CUS999 does not exist in Customer_Master; name and country (Lotus Department Store, Singapore) match CUS012.	Flag as Unknown customer.	Lookup =XLOOKUP(ID,Master[ID],Master[ID],"Not found") or =ISNA(MATCH(ID,Master[ID],0)).
25	715	ORD10714	Product_ID	Unmatched ID	Unmatched ID	PR999	PR008	Re-key from Product_Name via Product_Master	PR999 does not exist in Product_Master; name, category, fabric, construction and GSM all match PR008.	Flag as Unknown product.	Lookup =XLOOKUP(ID,Master[ID],Master[ID],"Not found") or =ISNA(MATCH(ID,Master[ID],0)).
26	816	ORD10815	Factory_ID	Unmatched ID	Unmatched ID	F99	F04	Re-key from Factory name via Factory_Master	F99 does not exist in Factory_Master; factory name matches F04.	Flag as Unknown factory.	Lookup =XLOOKUP(ID,Master[ID],Master[ID],"Not found") or =ISNA(MATCH(ID,Master[ID],0)).
27	917	ORD10916	Salesperson_ID	Unmatched ID	Unmatched ID	S99	S04	Re-key from the Salesperson name via Salesperson_Master	S99 does not exist in Salesperson_Master; the row name (T. Hossain) matches exactly one salesperson.	Flag as Unknown and exclude from salesperson views.	Lookup =XLOOKUP(ID,Master[ID],Master[ID],"Not found") or =ISNA(MATCH(ID,Master[ID],0)).
28	107	ORD10106	Order_Date	Mixed date format	Mixed date format (text) + conflicting value	2026/02/14	2026-08-06	Convert text to a real date; value reconstructed as Actual_Ship_Date - Lead_Time_Days (2026-10-06 - 61)	Stored as text '2026/02/14' (yyyy/mm/dd). Taken literally it gives a 227-day promise window and 234-day lead time (all other orders: 12-65 and 19-82 days) and contradicts Lead_Time_Days=61, which equals Ship - Order on all 990 complete rows.	Convert literally to 2026-02-14 (moves one order from Aug to Feb).	Test =ISNUMBER(date): FALSE marks text dates; convert with DATEVALUE / DATE(LEFT,MID,RIGHT) / Text to Columns; validate against Lead_Time_Days.
29	511	ORD10510	Promised_Date	Mixed date format	Mixed date format (text)	15-06-2026	2026-06-15	Convert dd-mm-yyyy text to a real date (DATE(RIGHT,MID,LEFT) or Text to Columns DMY)	Stored as text '15-06-2026'. Day 15 cannot be a month, so it is unambiguous: 15-Jun-2026. Kept as stated; note it gives the shortest promise window (12 days) and 49 days late.	-	Test =ISNUMBER(date): FALSE marks text dates; convert with DATEVALUE / DATE(LEFT,MID,RIGHT) / Text to Columns; validate against Lead_Time_Days.
30	808	ORD10807	Actual_Ship_Date	Mixed date format	Mixed date format (text) + conflicting value	2026-08-31	2026-04-12	Convert text to a real date; value reconstructed as Order_Date + Lead_Time_Days (2026-02-08 + 63)	Stored as text '2026-08-31'. Taken literally it gives 158 days late and a 204-day lead time (max elsewhere 18 and 82) and contradicts Lead_Time_Days=63.	Convert literally to 2026-08-31 (order is Late either way).	Test =ISNUMBER(date): FALSE marks text dates; convert with DATEVALUE / DATE(LEFT,MID,RIGHT) / Text to Columns; validate against Lead_Time_Days.

Task 1 - Cleaning Log, Reconciliation & Validation Checks

A. Issues found by category

Issue category	Count	Fields affected	How detected (Excel)	Rule applied	Records / cells
Irrelevant record	1	Entire row	Filter Order_ID: text that is not an ORD##### ID; all other fields blank (COUNTBLANK on the row = 30).	Delete the row (not an order line).	(none)
Duplicate record	5	Entire row	Helper =COUNTIF($A$2:A2,A2)>1 flags repeat IDs; compare all fields (COUNTIFS) or Data > Remove Duplicates on all columns.	Keep first occurrence; delete exact copies (all 31 fields identical).	ORD10042, ORD10177, ORD10388, ORD10721, ORD10902
Missing value	7	Actual_Ship_Date, Customer_Name, Defect_Qty, Lead_Time_Days, Salesperson_ID, Season, Unit_Price_USD	Filter (Blanks) on each column, =COUNTBLANK(range), or Go To Special > Blanks.	Recover from masters or related fields where determined (names, IDs, price, ship date, lead time); otherwise blank/"Unknown" and exclude from the affected metric only.	ORD10068, ORD10122, ORD10215, ORD10317, ORD10504, ORD10645, ORD10812
Invalid / anomalous value	5	Delivered_Qty, Lead_Time_Days, Order_Qty, Revenue_USD, Unit_Price_USD	Rule flag =IF(OR(Qty<=0,Price<=0,Defect>Qty),"Check","OK") ; sort descending (spots 9,999,999 and 999); Revenue vs Qty x Price check.	Recover from the arithmetic identities (Revenue = Qty x Price; Lead time = Ship - Order); otherwise treat as unknown and exclude from that metric.	ORD10334, ORD10445, ORD10556, ORD10778, ORD10889
Inconsistent label / format	5	Channel, Quality_Status, Salesperson_ID	Filter drop-down or =UNIQUE(range) lists the variants; EXACT/PROPER/TRIM; Find & Replace.	Standardise to the master spelling/case; re-key the malformed salesperson ID from the name.	ORD10092, ORD10156, ORD10267, ORD10478, ORD10700
Unmatched ID	4	Customer_ID, Factory_ID, Product_ID, Salesperson_ID	Lookup =XLOOKUP(ID,Master[ID],Master[ID],"Not found") or =ISNA(MATCH(ID,Master[ID],0)).	Re-key from the name on the same row (every name matches exactly one master record).	ORD10613, ORD10714, ORD10815, ORD10916
Mixed date format	3	Actual_Ship_Date, Order_Date, Promised_Date	Test =ISNUMBER(date): FALSE marks text dates; convert with DATEVALUE / DATE(LEFT,MID,RIGHT) / Text to Columns; validate against Lead_Time_Days.	Convert text to real dates; where the literal value contradicts Lead_Time_Days, reconstruct from the identity.	ORD10106, ORD10510, ORD10807
Total issues logged	**30**				

B. Row and value reconciliation (raw -> clean)

Row bridge	Value	Note
Rows in Raw_Orders (excl. header)	1,001	Includes one non-data notice row
less: non-data / irrelevant rows	-1	Raw row 1002
less: exact duplicate rows	-5	Raw rows 44, 179, 390, 723, 904
Equals: expected clean order lines	995	
Order lines in Clean_Orders	995	
Check	OK - reconciled	

Value bridge	USD / units	Note
Raw revenue (sum of Raw_Orders!V2:V1001)	30,242,249.68	As supplied
less: revenue on duplicate rows	-132,314.31	Double-counted order lines
less: placeholder correction (ORD10556)	-9,985,385.30	9,999,999 replaced by Qty x Price
Equals: clean revenue (bridge)	20,124,550.07	
Revenue in Clean_Orders	20,124,550.07	
Check	OK - reconciled	
Raw units (sum of Raw_Orders!T2:T1001)	2,599,233.00	As supplied
less: units on duplicate rows	-13,275.00	
plus: quantity correction (ORD10334)	4,647.00	-250 replaced by Revenue / Price
Equals: clean units (bridge)	2,590,605.00	
Units in Clean_Orders	2,590,605.00	
Check	OK - reconciled	
Raw gross margin (sum of Raw_Orders!AD2:AD10(	11,386,779.31	
less: margin on duplicate rows	-70,070.92	
Equals: clean gross margin	11,316,708.39	
Check vs Clean_Orders	OK - reconciled	

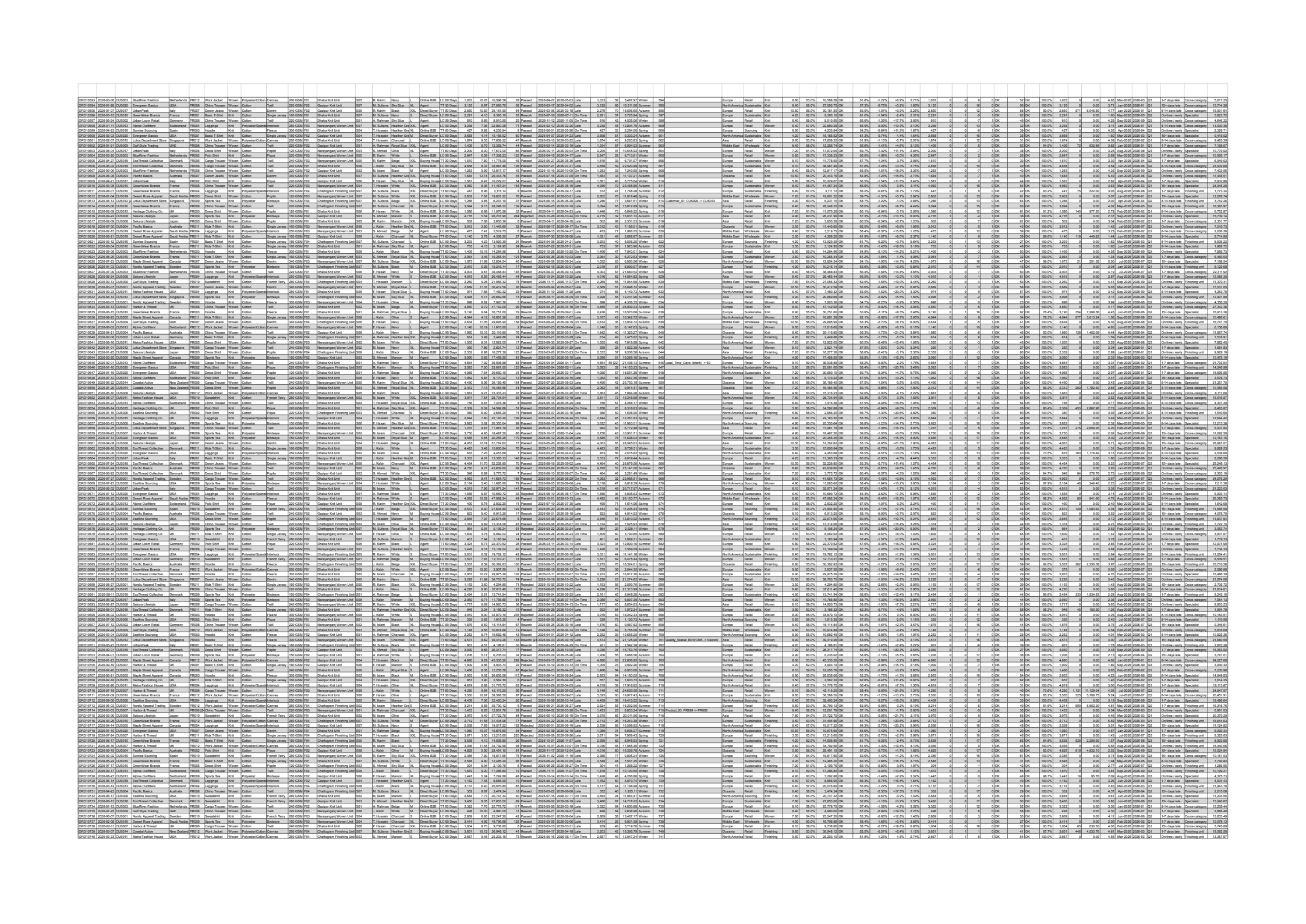
C. Validation checks on Clean_Orders (all should be 0 / OK)

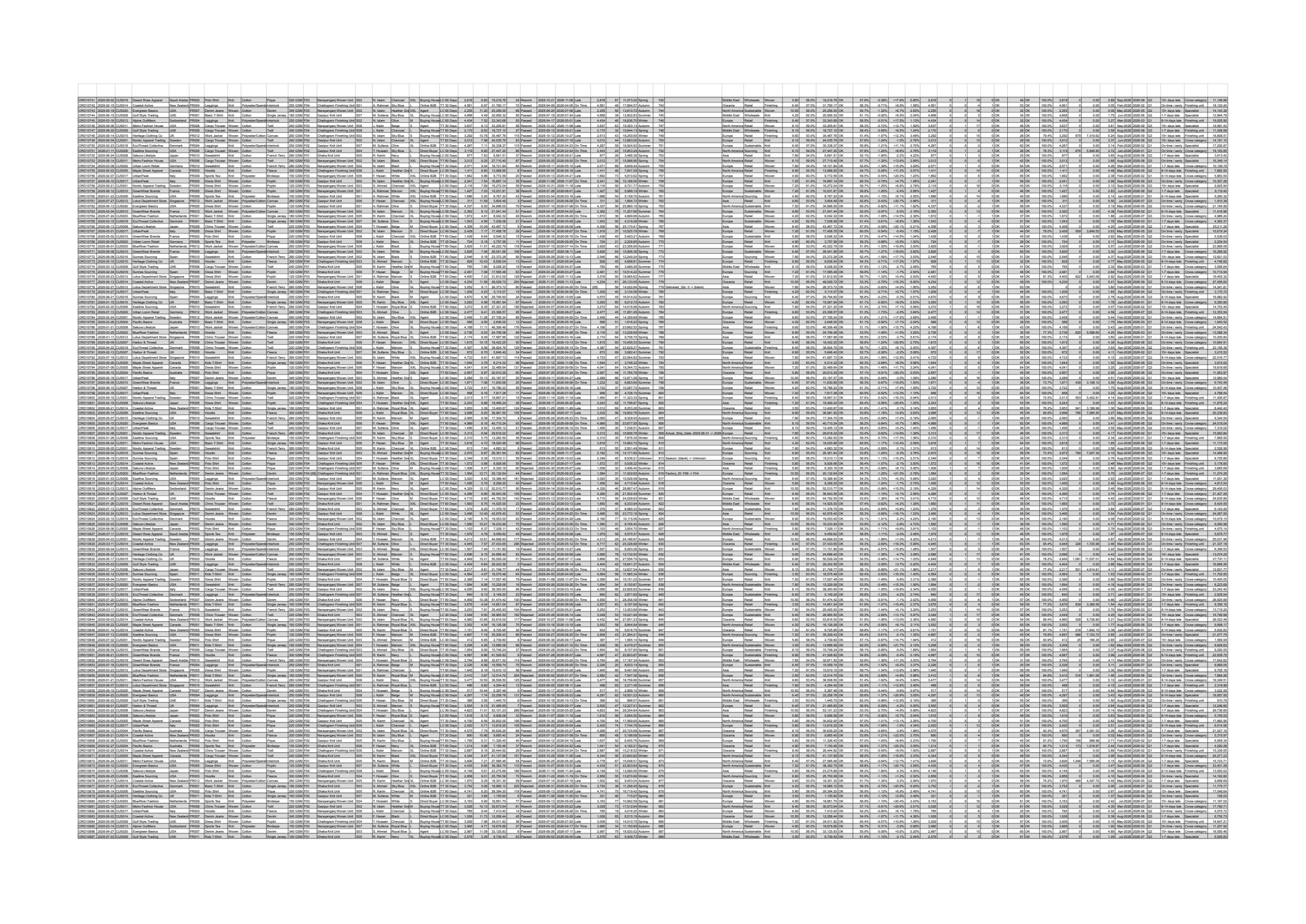
Check	Result	Note
Revenue = Order_Qty x Unit_Price (rows failing)	0	Helper column Revenue_Check
Delivery_Status agrees with Actual vs Promised dat	0	Helper column Status_Check (data dictionary: "validate against dates")
Lead_Time_Days = Actual_Ship_Date - Order_Date	0	Helper column Lead_Time_Check
Defect_Qty > Order_Qty	0	Data dictionary rule
Delivered_Qty > Order_Qty	0	
Order_Qty <= 0 or Unit_Price <= 0	0	
Customer_ID not in Customer_Master	0	
Product_ID not in Product_Master	0	
Factory_ID not in Factory_Master	0	
Salesperson_ID not in Salesperson_Master	0	
Duplicate Order_IDs remaining	0	
Text (non-date) values in date columns	0	
Distinct Channel labels (expect 4)	4	Agent, Buying House, Direct Buyer, Online B2B
Distinct Quality_Status labels (expect 3)	3	Passed, Rework, Rejected
Production_Data: duplicate Production_IDs	0	Supporting dataset is clean
Production_Data: Defect_Rate_% differs from Defe	0	Agrees within rounding (+/-0.005 pts) on all 300 batches
Monthly_Targets: rows (expect 9 months x 8 salesp	72	

D. Reviewed and deliberately not changed

Item	Count / value	Decision
Factory vs product category	372	Orders made in a factory whose type differs from the product category (e.g. woven garments at knit units). Defect and on-time performance is the same as specialist allocation (see Ops_Analysis), so treated as valid cross-loading, not an error.
Season vs order month	1	All four seasons occur in every order month, so Season is a buyer-season label, not derived from Order_Date. Values kept; the one blank (count shown) set to "Unknown".
Partial deliveries (Delivered < Ordered)	170	Short shipments of 1-22% are a genuine business pattern (fill-rate KPI), not data errors.
Text date ORD10510 Promised_Date	49	Converted literally to 15-Jun-2026 (unambiguous). It yields the shortest promise window (12 days) and 49 days late - flag for review; no evidence to change it.







Task 3 - Sales & Commercial Analysis

Jump to: KPIs | Product | Category | Customer | Region | Salesperson | Month | Channel/Terms/Season

A. Headline commercial KPIs

Metric	Value	How it is calculated
Order lines (after cleaning)	**995**	*1,000 raw order rows - 5 duplicates (+1 non-data row removed)*
Units ordered	**2,590,605**	*SUM(Order_Qty)*
Revenue (USD)	**$20,124,550**	*SUM(Revenue_USD) after fixing the 9,999,999 placeholder*
Gross margin (USD)	**$11,316,708**	*SUM(Gross_Margin_USD)*
Gross margin %	**56.2%**	*Total margin / total revenue (not an average of row %)*
Target margin % (revenue-weighted)	**57.4%**	*Product_Master Target_Margin_Rate weighted by revenue*
Margin gap vs target (pp)	**-1.13%**	*Actual GM% - target GM%*
Margin shortfall vs target (USD)	**$227,507**	*Margin that target rates would have produced, minus actual*
Orders below their product target margi	**995**	*Helper column Margin_Gap_vs_Target < 0*
Average selling price (USD per unit)	**$7.77**	*Revenue / units*
Average order value (USD)	**$20,226**	*Revenue / order lines*
Price premium vs standard price	**+8.2%**	*Actual revenue vs units x Product_Master standard price*
Revenue target Jan-Sep (USD)	**$21,110,246**	*SUM(Monthly_Targets[Target_Revenue_USD])*
Target attainment %	**95.3%**	*Revenue / target*
Revenue gap to target (USD)	**-$985,696**	*Negative = shortfall*
Salesperson-months at or above target	**37**	*Target_Analysis column "Met target?"*
Salesperson-months in plan	**72**	*9 months x 8 salespeople*
Woven share of revenue	**54.1%**	*Woven revenue / total*
Woven share of order lines	**43.9%**	*Woven orders / total*
Europe share of revenue	**48.6%**	*Region via Customer_Master lookup*
Largest customer	**Sakura Lifestyle**	*From customer table (section D)*
Largest customer share of revenue	**6.5%**	*Concentration check*
Top-5 customers share of revenue	**30.9%**	*Pareto check*
Top-4 products share of revenue	**45.2%**	*From product table (section B)*

B. Product performance (sorted by revenue)

Product_ID	Product	Category	Orders	Units	Revenue (USD)	Rev. share	Rank	Gross margin (USD)	GM %	Target GM %	GM gap vs target	Avg price	Std price	Price vs std	Defect %	Rejected %	On-time %
PR007	Denim Jeans	Woven	82	216,287	$2,480,669	12.3%	1	$1,360,731	54.9%	56.0%	-1.15%	$11.47	$10.50	+9.2%	2.54%	11.0%	35.4%
PR012	Work Jacket	Woven	91	233,679	$2,394,298	11.9%	2	$1,239,814	51.8%	53.0%	-1.22%	$10.25	$9.60	+6.7%	2.69%	7.7%	34.1%
PR008	Chino Trouser	Woven	91	228,088	$2,116,932	10.5%	3	$1,202,420	56.8%	58.0%	-1.20%	$9.28	$8.40	+10.5%	2.68%	9.9%	28.6%
PR003	Hoodie	Knit	86	220,003	$2,100,395	10.4%	4	$1,132,126	53.9%	55.0%	-1.10%	$9.55	$8.90	+7.3%	2.46%	7.0%	33.7%
PR006	Cargo Trouser	Woven	82	225,570	$1,958,031	9.7%	5	$1,132,619	57.8%	59.0%	-1.16%	$8.68	$8.10	+7.2%	2.55%	7.3%	34.1%
PR005	Dress Shirt	Woven	91	249,291	$1,945,533	9.7%	6	$1,166,162	59.9%	61.0%	-1.06%	$7.80	$7.20	+8.4%	2.35%	7.7%	31.9%
PR004	Leggings	Knit	85	226,307	$1,556,478	7.7%	7	$870,777	55.9%	57.0%	-1.05%	$6.88	$6.40	+7.5%	2.32%	8.2%	34.1%
PR010	Sweatshirt	Knit	69	164,226	$1,374,774	6.8%	8	$727,646	52.9%	54.0%	-1.07%	$8.37	$7.80	+7.3%	2.37%	7.2%	29.0%
PR009	Sports Tee	Knit	87	238,403	$1,283,442	6.4%	9	$755,690	58.9%	60.0%	-1.12%	$5.38	$4.90	+9.9%	2.48%	5.7%	31.0%
PR002	Polo Shirt	Knit	81	204,314	$1,281,787	6.4%	10	$729,115	56.9%	58.0%	-1.12%	$6.27	$5.80	+8.2%	2.51%	8.6%	25.9%
PR001	Basic T-Shirt	Knit	84	228,055	$1,035,362	5.1%	11	$630,568	60.9%	62.0%	-1.10%	$4.54	$4.20	+8.1%	2.44%	6.0%	25.0%
PR011	Kids T-Shirt	Knit	66	156,382	$596,850	3.0%	12	$369,041	61.8%	63.0%	-1.17%	$3.82	$3.50	+9.0%	2.62%	6.1%	30.3%
Total			**995**	**2,590,605**	**$20,124,550**	**100.0%**		**$11,316,708**	**56.2%**	**57.4%**	**-1.13%**	**$7.77**		**+8.2%**	**2.50%**	**7.7%**	**31.2%**

C. Category (Knit vs Woven)

Task 3 - Sales & Commercial Analysis

Jump to: KPIs | Product | Category | Customer | Region | Salesperson | Month | Channel/Terms/Season

Category	Orders	Units	Revenue (USD)	Rev. share	GM %	Avg price	Defect %	Rejected %	On-time %
Woven	437	1,152,915	$10,895,462	54.1%	56.0%	$9.45	2.56%	8.7%	32.7%
Knit	558	1,437,690	$9,229,088	45.9%	56.5%	$6.42	2.45%	7.0%	29.9%
Total	**995**	**2,590,605**	**$20,124,550**	**100.0%**	**56.2%**	**$7.77**	**2.50%**	**7.7%**	**31.2%**

D. Customer performance (sorted by revenue, with Pareto)

Customer_ID	Customer	Country	Region	Type	Orders	Revenue (USD)	Rev. share	Cum. share	Rank	GM %	On-time %	Rejected %
CUS008	Sakura Lifestyle	Japan	Asia	Retail	58	$1,316,414	6.5%	6.5%	1	56.2%	39.7%	8.6%
CUS006	Eastline Sourcing	USA	North America	Sourcing	64	$1,313,655	6.5%	13.1%	2	56.5%	23.4%	7.8%
CUS018	EcoThread Collective	Denmark	Europe	Sustainable	63	$1,244,492	6.2%	19.3%	3	55.8%	34.9%	9.5%
CUS017	UrbanPeak	Italy	Europe	Retail	62	$1,219,988	6.1%	25.3%	4	56.4%	37.1%	3.2%
CUS007	Harbor & Thread	UK	Europe	Retail	59	$1,126,369	5.6%	30.9%	5	57.2%	30.5%	11.9%
CUS020	Evergreen Basics	USA	North America	Sustainable	52	$1,121,041	5.6%	36.5%	6	56.1%	30.8%	3.8%
CUS010	GreenWear Brands	France	Europe	Sustainable	59	$1,110,538	5.5%	42.0%	7	56.5%	27.1%	3.4%
CUS009	Gulf Style Trading	UAE	Middle East	Wholesale	51	$1,103,115	5.5%	47.5%	8	56.3%	43.1%	7.8%
CUS014	Coastal Active	New Zealand	Oceania	Retail	45	$1,045,210	5.2%	52.7%	9	55.6%	37.8%	8.9%
CUS001	Nordic Apparel Trading	Sweden	Europe	Retail	51	$1,012,468	5.0%	57.7%	10	55.6%	29.4%	7.8%
CUS012	Lotus Department Store	Singapore	Asia	Retail	52	$986,764	4.9%	62.6%	11	56.7%	36.5%	3.8%
CUS019	Desert Rose Apparel	Saudi Arabia	Middle East	Wholesale	43	$941,049	4.7%	67.3%	12	56.0%	20.9%	7.0%
CUS013	Alpine Outfitters	Switzerland	Europe	Retail	45	$940,352	4.7%	72.0%	13	55.9%	28.9%	6.7%
CUS005	Maple Street Apparel	Canada	North America	Retail	41	$862,965	4.3%	76.2%	14	56.9%	26.8%	22.0%
CUS003	BlueRiver Fashion	Netherlands	Europe	Retail	41	$836,998	4.2%	80.4%	15	55.7%	24.4%	7.3%
CUS004	Pacific Basics	Australia	Oceania	Retail	37	$827,638	4.1%	84.5%	16	56.6%	18.9%	5.4%
CUS016	Sunrise Sourcing	Spain	Europe	Sourcing	53	$824,472	4.1%	88.6%	17	57.0%	30.2%	5.7%
CUS011	Metro Fashion House	USA	North America	Retail	41	$818,062	4.1%	92.7%	18	56.1%	34.1%	4.9%
CUS015	Heritage Clothing Co	UK	Europe	Retail	39	$757,449	3.8%	96.4%	19	56.0%	30.8%	15.4%
CUS002	Urban Loom Retail	Germany	Europe	Retail	39	$715,513	3.6%	100.0%	20	55.4%	30.8%	7.7%
Total					**995**	**$20,124,550**	**100.0%**			**56.2%**	**31.2%**	**7.7%**

E. Market view: region, country and customer type

Region	Customers	Orders	Revenue (USD)	Rev. share	GM %	On-time %	Rejected %	Defect %
Europe	10	511	$9,788,638	48.6%	56.2%	30.7%	7.6%	2.47%
North America	4	198	$4,115,723	20.5%	56.4%	28.3%	9.1%	2.59%
Asia	2	110	$2,303,178	11.4%	56.4%	38.2%	6.4%	2.62%
Middle East	2	94	$2,044,164	10.2%	56.2%	33.0%	7.4%	2.39%
Oceania	2	82	$1,872,848	9.3%	56.1%	29.3%	7.3%	2.40%
Total	**20**	**995**	**$20,124,550**	**100.0%**	**56.2%**	**31.2%**	**7.7%**	**2.50%**

Country	Region	Orders	Revenue (USD)	Rev. share	Rank	On-time %	Rejected %
USA	North America	157	$3,252,758	16.2%	1	28.7%	5.7%
UK	Europe	98	$1,883,817	9.4%	2	30.6%	13.3%
Japan	Asia	58	$1,316,414	6.5%	3	39.7%	8.6%

Task 3 - Sales & Commercial Analysis

Jump to: KPIs | Product | Category | Customer | Region | Salesperson | Month | Channel/Terms/Season

Denmark	Europe	63	$1,244,492	6.2%	4	34.9%	9.5%											
Italy	Europe	62	$1,219,988	6.1%	5	37.1%	3.2%											
France	Europe	59	$1,110,538	5.5%	6	27.1%	3.4%											
UAE	Middle East	51	$1,103,115	5.5%	7	43.1%	7.8%											
New Zealand	Oceania	45	$1,045,210	5.2%	8	37.8%	8.9%											
Sweden	Europe	51	$1,012,468	5.0%	9	29.4%	7.8%											
Singapore	Asia	52	$986,764	4.9%	10	36.5%	3.8%											
Saudi Arabia	Middle East	43	$941,049	4.7%	11	20.9%	7.0%											
Switzerland	Europe	45	$940,352	4.7%	12	28.9%	6.7%											
Canada	North America	41	$862,965	4.3%	13	26.8%	22.0%											
Netherlands	Europe	41	$836,998	4.2%	14	24.4%	7.3%											
Australia	Oceania	37	$827,638	4.1%	15	18.9%	5.4%											
Spain	Europe	53	$824,472	4.1%	16	30.2%	5.7%											
Germany	Europe	39	$715,513	3.6%	17	30.8%	7.7%											
Total		**995**	**$20,124,550**	**100.0%**		**31.2%**	**7.7%**											

Customer type	Orders	Revenue (USD)	Rev. share	GM %	On-time %	Rejected %	Defect %
Retail	610	$12,466,189	61.9%	56.2%	31.8%	8.5%	2.58%
Sustainable	174	$3,476,070	17.3%	56.1%	31.0%	5.7%	2.35%
Sourcing	117	$2,138,126	10.6%	56.7%	26.5%	6.8%	2.41%
Wholesale	94	$2,044,164	10.2%	56.2%	33.0%	7.4%	2.39%
Total	**995**	**$20,124,550**	**100.0%**	**56.2%**	**31.2%**	**7.7%**	**2.50%**

F. Salesperson performance vs target (Jan-Sep)

SP_ID	Salesperson	Home region	Orders	Revenue (USD)	Rev. share	GM %	Target (USD)	Attainment %	Variance (USD)	Months on target	On-time %	Rejected %
S01	A. Rahman	Europe	115	$2,241,832	11.1%	56.5%	$2,828,027	79.3%	-$586,195	4	29.6%	8.7%
S02	N. Islam	North America	134	$2,965,202	14.7%	55.6%	$2,773,189	106.9%	+$192,013	5	36.6%	4.5%
S03	S. Ahmed	Asia	111	$2,279,921	11.3%	56.0%	$2,582,183	88.3%	-$302,262	4	28.8%	9.0%
S04	T. Hossain	Middle East	121	$2,680,082	13.3%	56.2%	$2,803,963	95.6%	-$123,881	4	28.9%	8.3%
S05	R. Karim	Oceania	129	$2,431,723	12.1%	56.5%	$2,555,429	95.2%	-$123,706	5	28.7%	8.5%
S06	F. Hasan	Europe	121	$2,316,217	11.5%	56.4%	$2,514,639	92.1%	-$198,422	5	32.2%	9.1%
S07	M. Sultana	North America	124	$2,205,875	11.0%	56.3%	$2,510,005	87.9%	-$304,130	4	33.1%	10.5%
S08	J. Kabir	Europe	140	$3,003,700	14.9%	56.4%	$2,542,811	118.1%	+$460,889	6	30.7%	4.3%
Total			**995**	**$20,124,550**	**100.0%**	**56.2%**	**$21,110,24**	**95.3%**	**-$985,696**	**37**	**31.2%**	**7.7%**

G. Monthly trend vs target

Month	Month start	Key	Orders	Units	Revenue (USD)	MoM change	Gross margin (USD)	GM %	Target (USD)	Variance (USD)	Attainment %	On-time %	Defect %	Rejected %
Jan 2026	Jan-2026	2026-01	108	290,189	$2,260,036	-	$1,272,730	56.3%	$2,043,778	+$216,258	110.6%	28.7%	2.66%	11.1%
Feb 2026	Feb-2026	2026-02	124	302,552	$2,360,607	+4.4%	$1,328,861	56.3%	$2,429,595	-$68,988	97.2%	32.3%	2.55%	8.9%
Mar 2026	Mar-2026	2026-03	116	326,171	$2,692,551	+14.1%	$1,500,199	55.7%	$2,307,022	+$385,529	116.7%	31.9%	2.51%	7.8%
Apr 2026	Apr-2026	2026-04	101	238,314	$1,871,269	-30.5%	$1,049,198	56.1%	$2,421,701	-$550,432	77.3%	22.8%	2.36%	7.9%
May 2026	May-2026	2026-05	96	230,768	$1,794,070	-4.1%	$1,012,779	56.5%	$2,548,334	-$754,264	70.4%	33.3%	2.08%	3.1%
Jun 2026	Jun-2026	2026-06	130	343,462	$2,543,161	+41.8%	$1,433,653	56.4%	$2,372,622	+$170,539	107.2%	27.7%	2.37%	4.6%

Task 3 - Sales & Commercial Analysis

Jump to: KPIs | Product | Category | Customer | Region | Salesperson | Month | Channel/Terms/Season

Jul 2026	Jul-2026	2026-07	98	268,159	$2,116,724	-16.8%	$1,188,300	56.1%	$2,185,432	-$68,708	96.9%	33.7%	2.75%	9.2%			
Aug 2026	Aug-2026	2026-08	115	294,086	$2,244,467	+6.0%	$1,263,507	56.3%	$2,564,586	-$320,119	87.5%	35.7%	2.65%	11.3%			
Sep 2026	Sep-2026	2026-09	107	296,904	$2,241,665	-0.1%	$1,267,481	56.5%	$2,237,176	+$4,489	100.2%	34.6%	2.50%	5.6%			
Total			**995**	**2,590,605**	**$20,124,550**		**$11,316,70**	**56.2%**	**$21,110,246**	**-$985,696**	**95.3%**	**31.2%**	**2.50%**	**7.7%**			

H. Channel, payment terms and season

Channel	Orders	Revenue (USD)	Rev. share	GM %	On-time %	Rejected %	Defect %
Agent	257	$5,363,063	26.6%	56.0%	27.2%	10.1%	2.46%
Direct Buyer	246	$4,956,965	24.6%	56.4%	31.3%	6.5%	2.51%
Buying House	253	$4,949,932	24.6%	56.0%	31.2%	7.9%	2.50%
Online B2B	239	$4,854,591	24.1%	56.4%	35.1%	6.3%	2.54%
Total	**995**	**$20,124,550**	**100.0%**	**56.2%**	**31.2%**	**7.7%**	**2.50%**

Payment terms	Orders	Revenue (USD)	Rev. share	GM %	On-time %	Rejected %	Defect %
LC 60 Days	262	$5,436,181	27.0%	56.1%	30.9%	7.3%	2.53%
TT 30 Days	250	$4,948,953	24.6%	56.4%	31.6%	6.4%	2.41%
TT 60 Days	234	$4,887,551	24.3%	56.3%	32.5%	7.7%	2.52%
LC 90 Days	249	$4,851,865	24.1%	56.2%	29.7%	9.6%	2.54%
Total	**995**	**$20,124,550**	**100.0%**	**56.2%**	**31.2%**	**7.7%**	**2.50%**

Season	Orders	Revenue (USD)	Rev. share	GM %	On-time %	Rejected %	Defect %
Spring	274	$5,668,525	28.2%	56.3%	33.9%	9.9%	2.69%
Autumn	266	$5,452,655	27.1%	56.2%	31.2%	5.6%	2.41%
Winter	242	$4,776,541	23.7%	55.9%	32.6%	6.6%	2.55%
Summer	212	$4,211,819	20.9%	56.5%	25.9%	9.0%	2.31%
Unknown	1	$15,010	0.1%	56.9%	0.0%	0.0%	2.51%
Total	**995**	**$20,124,550**	**100.0%**	**56.2%**	**31.2%**	**7.7%**	**2.50%**

Task 3 - Quality, Delivery & Production Analysis

Jump to: KPIs | Factory | Quality mix | Lateness | Product ops | Allocation | Production

A. Headline operations KPIs

Metric	Value	How it is calculated
Defect rate (defective units / units inspected)	**2.50%**	*Order with missing Defect_Qty excluded from the denominator*
Defective units	**64,756**	*SUM(Defect_Qty)*
Estimated cost of defective units (USD)	**$220,165**	*Defective units x average unit cost*
Rejected orders	**77**	*Quality_Status = Rejected*
Rejected orders %	**7.7%**	
Revenue on rejected orders (USD)	**$1,583,938**	*Revenue exposed to claims / chargebacks*
Rework orders	**157**	*Quality_Status = Rework*
Rework orders %	**15.8%**	
Revenue on rework orders (USD)	**$2,856,165**	
Defect rate - rejected lots	**5.98%**	*Rejected lots cluster at ~6% defects*
Defect rate - passed lots	**2.18%**	
On-time orders	**310**	*Actual_Ship_Date <= Promised_Date*
On-time delivery %	**31.2%**	*Same result from Delivery_Status (validated: 0 conflicts)*
Late orders	**685**	
Revenue shipped late (USD)	**$13,875,460**	
Share of revenue shipped late	**68.9%**	
Average days late (late orders only)	**9.6**	*Days_Late = Actual_Ship - Promised*
Maximum days late	**49**	
Orders more than 7 days late	**424**	
Share of orders more than 7 days late	**42.6%**	
Average promise window (order to promised date, day	**45.1**	*What customers are quoted*
Average actual lead time (order to ship, days)	**50.6**	*What the factories deliver*
Lead-time gap (actual - promised, days)	**5.5**	*Positive = promises are shorter than real lead time*
Fill rate (delivered / ordered units)	**97.2%**	*Order with invalid 0 delivered excluded*
Short-shipped orders	**170**	*Delivered_Qty < Order_Qty*
Units not delivered	**73,580**	
Value of units not delivered (USD)	**$578,631**	*Short units x unit price*
Production plan attainment (produced / planned)	**98.2%**	*Production_Data, 300 batches*
Production batches below plan	**62.0%**	
Production defect rate (defects / produced)	**3.02%**	*In-process defects (before shipment)*
Rework recovery (rework / defects)	**26.7%**	*Share of defects recovered by rework*

B. Factory scorecard - orders (Clean_Orders) + production (Production_Data)

Factory_ID	Factory	Type	Orders	Revenue (USD)	Defect %	Rejected %	Rework %	On-time %	Avg days late	Fill rate	Avg lead time	Planned (prod.)	Produced	Plan attainment	Prod. defect %	Rework recovery
F01	Dhaka Knit Unit	Knit	247	$4,951,191	2.45%	7.3%	14.2%	29.1%	9.8	97.0%	50.9	289,435	285,107	98.5%	2.81%	26.5%
F02	Gazipur Knit Unit	Knit	239	$4,869,495	2.53%	8.8%	15.5%	31.0%	9.5	97.2%	50.5	241,885	237,206	98.1%	2.99%	25.9%
F03	Narayanganj Woven Unit	Woven	272	$5,545,906	2.43%	6.3%	16.9%	31.6%	9.7	97.1%	49.9	286,959	281,470	98.1%	3.03%	25.9%
F04	Chattogram Finishing Unit	Finishing	237	$4,757,958	2.61%	8.9%	16.5%	32.9%	9.4	97.3%	51.1	274,175	269,531	98.3%	3.24%	28.3%
Total			**995**	**$20,124,550**	**2.50%**	**7.7%**	**15.8%**	**31.2%**	**9.6**	**97.2%**	**50.6**	**1,092,454**	**1,073,314**	**98.2%**	**3.02%**	**26.7%**

C. Quality outcome mix

Quality status	Orders	% of orders	Revenue (USD)	% of revenue	Defect rate	GM %
Passed	761	76.5%	$15,684,447	77.9%	2.18%	56.4%

Task 3 - Quality, Delivery & Production Analysis

Jump to: KPIs | Factory | Quality mix | Lateness | Product ops | Allocation | Production

Rework	157	15.8%	$2,856,165	14.2%	2.37%	56.2%	
Rejected	77	7.7%	$1,583,938	7.9%	5.98%	54.6%	
Total	**995**	**100.0%**	**$20,124,550**	**100.0%**	**2.50%**	**56.2%**	

D. Delivery lateness distribution

Delivery bucket	Orders	% of orders	Revenue (USD)	% of revenue
On time / early	310	31.2%	$6,249,090	31.1%
1-7 days late	261	26.2%	$5,173,889	25.7%
8-14 days late	279	28.0%	$5,562,189	27.6%
15+ days late	145	14.6%	$3,139,382	15.6%
Total	**995**	**100.0%**	**$20,124,550**	**100.0%**

E. Quality and delivery by product

Product_ID	Product	Orders	Defect %	Rejected %	Rework %	On-time %	Avg days late	Fill rate	Prod. defect %
PR007	Denim Jeans	82	2.54%	11.0%	13.4%	35.4%	10.4	97.5%	2.95%
PR012	Work Jacket	91	2.69%	7.7%	20.9%	34.1%	8.7	97.2%	2.96%
PR008	Chino Trouser	91	2.68%	9.9%	14.3%	28.6%	9.2	97.1%	3.16%
PR003	Hoodie	86	2.46%	7.0%	16.3%	33.7%	11.5	94.8%	2.39%
PR006	Cargo Trouser	82	2.55%	7.3%	11.0%	34.1%	9.7	96.4%	2.87%
PR005	Dress Shirt	91	2.35%	7.7%	8.8%	31.9%	9.8	97.5%	2.80%
PR004	Leggings	85	2.32%	8.2%	15.3%	34.1%	8.6	98.2%	3.11%
PR010	Sweatshirt	69	2.37%	7.2%	13.0%	29.0%	9.9	99.3%	2.98%
PR009	Sports Tee	87	2.48%	5.7%	14.9%	31.0%	9.7	98.3%	3.21%
PR002	Polo Shirt	81	2.51%	8.6%	17.3%	25.9%	10.0	97.0%	3.40%
PR001	Basic T-Shirt	84	2.44%	6.0%	20.2%	25.0%	9.3	97.5%	2.86%
PR011	Kids T-Shirt	66	2.62%	6.1%	25.8%	30.3%	8.5	94.8%	3.19%
Total		**995**	**2.50%**	**7.7%**	**15.8%**	**31.2%**	**9.6**	**97.2%**	**3.02%**

F. Factory allocation check (specialist vs cross-category)

Allocation	Orders	Revenue (USD)	Defect %	Rejected %	On-time %	GM %
Specialist	386	$7,577,691	2.46%	7.0%	28.2%	56.2%
Cross-category	372	$7,788,901	2.47%	7.8%	33.1%	56.3%
Finishing unit	237	$4,757,958	2.61%	8.9%	32.9%	56.2%
Total	**995**	**$20,124,550**	**2.50%**	**7.7%**	**31.2%**	**56.2%**

Specialist = product category matches factory type (Knit at a knit unit, Woven at the woven unit). No performance penalty from cross-loading, so it is not treated as a data error.

G. Production_Data: process, factory x process and monthly view

Process	Batches	Planned	Produced	Plan attainment	Defect %	Rework recovery
Cutting	72	274,353	269,163	98.1%	2.88%	26.1%
Sewing	71	249,671	246,842	98.9%	2.97%	25.8%
Finishing	89	301,849	297,369	98.5%	3.21%	27.1%

Task 3 - Quality, Delivery & Production Analysis

Jump to: KPIs | Factory | Quality mix | Lateness | Product ops | Allocation | Production

Packing	68	266,581	259,940	97.5%	2.98%	27.6%
Total	**300**	**1,092,454**	**1,073,314**	**98.2%**	**3.02%**	**26.7%**

Prod. defect % (factory x process)	Cutting	Sewing	Finishing	Packing	All processes
Dhaka Knit Unit	3.20%	2.89%	2.26%	2.98%	**2.81%**
Gazipur Knit Unit	2.94%	2.79%	3.50%	2.64%	**2.99%**
Narayanganj Woven Unit	2.66%	2.58%	3.72%	3.10%	**3.03%**
Chattogram Finishing Unit	2.79%	3.60%	3.42%	3.18%	**3.24%**

Production month	Batches	Planned	Produced	Plan attainment	Defect %
Jan-2026	27	75,779	73,728	97.3%	3.32%
Feb-2026	33	126,705	125,324	98.9%	3.60%
Mar-2026	36	123,365	120,503	97.7%	2.84%
Apr-2026	36	137,619	134,525	97.8%	2.61%
May-2026	26	94,223	92,744	98.4%	2.58%
Jun-2026	30	117,854	117,444	99.7%	3.28%
Jul-2026	36	133,697	131,615	98.4%	2.95%
Aug-2026	34	133,843	131,146	98.0%	2.93%
Sep-2026	42	149,369	146,285	97.9%	3.07%
Total	**300**	**1,092,454**	**1,073,314**	**98.2%**	**3.02%**

Task 2/3 - Target vs Actual by Salesperson and Month

Month	Month key	Salesperson_ID	Salesperson	Target (USD)	Actual (USD)	Orders	Variance (USD)	Met target?	Attainment %
Jan-2026	2026-01	S01	A. Rahman	$355,841	$364,988	18	+$9,147	Yes	102.6%
Jan-2026	2026-01	S02	N. Islam	$240,522	$260,162	11	+$19,640	Yes	108.2%
Jan-2026	2026-01	S03	S. Ahmed	$278,384	$167,061	8	-$111,323	No	60.0%
Jan-2026	2026-01	S04	T. Hossain	$190,689	$319,254	13	+$128,565	Yes	167.4%
Jan-2026	2026-01	S05	R. Karim	$255,628	$131,475	8	-$124,153	No	51.4%
Jan-2026	2026-01	S06	F. Hasan	$233,528	$375,249	17	+$141,721	Yes	160.7%
Jan-2026	2026-01	S07	M. Sultana	$266,164	$360,141	20	+$93,977	Yes	135.3%
Jan-2026	2026-01	S08	J. Kabir	$223,022	$281,705	13	+$58,683	Yes	126.3%
Feb-2026	2026-02	S01	A. Rahman	$257,458	$332,716	18	+$75,258	Yes	129.2%
Feb-2026	2026-02	S02	N. Islam	$332,665	$436,907	19	+$104,242	Yes	131.3%
Feb-2026	2026-02	S03	S. Ahmed	$389,027	$148,946	10	-$240,081	No	38.3%
Feb-2026	2026-02	S04	T. Hossain	$302,731	$284,067	13	-$18,664	No	93.8%
Feb-2026	2026-02	S05	R. Karim	$369,128	$269,788	18	-$99,340	No	73.1%
Feb-2026	2026-02	S06	F. Hasan	$184,367	$220,991	15	+$36,624	Yes	119.9%
Feb-2026	2026-02	S07	M. Sultana	$340,678	$277,461	15	-$63,217	No	81.4%
Feb-2026	2026-02	S08	J. Kabir	$253,541	$389,732	16	+$136,191	Yes	153.7%
Mar-2026	2026-03	S01	A. Rahman	$403,263	$318,909	13	-$84,354	No	79.1%
Mar-2026	2026-03	S02	N. Islam	$266,332	$390,916	15	+$124,584	Yes	146.8%
Mar-2026	2026-03	S03	S. Ahmed	$303,283	$321,514	14	+$18,231	Yes	106.0%
Mar-2026	2026-03	S04	T. Hossain	$416,911	$441,278	16	+$24,367	Yes	105.8%
Mar-2026	2026-03	S05	R. Karim	$321,060	$360,605	16	+$39,545	Yes	112.3%
Mar-2026	2026-03	S06	F. Hasan	$211,903	$199,048	7	-$12,855	No	93.9%
Mar-2026	2026-03	S07	M. Sultana	$194,856	$417,938	20	+$223,082	Yes	214.5%
Mar-2026	2026-03	S08	J. Kabir	$189,414	$242,344	15	+$52,930	Yes	127.9%
Apr-2026	2026-04	S01	A. Rahman	$349,587	$46,347	6	-$303,240	No	13.3%
Apr-2026	2026-04	S02	N. Islam	$300,233	$265,325	12	-$34,908	No	88.4%

Task 2/3 - Target vs Actual by Salesperson and Month

Month	Month key	Salesperson_ID	Salesperson	Target (USD)	Actual (USD)	Orders	Variance (USD)	Met target?	Attainment %
Apr-2026	2026-04	S03	S. Ahmed	$297,350	$296,557	13	-$793	No	99.7%
Apr-2026	2026-04	S04	T. Hossain	$268,486	$271,249	14	+$2,763	Yes	101.0%
Apr-2026	2026-04	S05	R. Karim	$182,105	$258,412	16	+$76,307	Yes	141.9%
Apr-2026	2026-04	S06	F. Hasan	$330,518	$276,329	15	-$54,189	No	83.6%
Apr-2026	2026-04	S07	M. Sultana	$322,387	$145,482	9	-$176,905	No	45.1%
Apr-2026	2026-04	S08	J. Kabir	$371,035	$311,567	16	-$59,468	No	84.0%
May-2026	2026-05	S01	A. Rahman	$191,082	$357,675	18	+$166,593	Yes	187.2%
May-2026	2026-05	S02	N. Islam	$301,067	$300,929	15	-$138	No	100.0%
May-2026	2026-05	S03	S. Ahmed	$363,637	$134,332	6	-$229,305	No	36.9%
May-2026	2026-05	S04	T. Hossain	$383,039	$223,591	11	-$159,448	No	58.4%
May-2026	2026-05	S05	R. Karim	$398,660	$144,407	10	-$254,253	No	36.2%
May-2026	2026-05	S06	F. Hasan	$409,075	$133,180	7	-$275,895	No	32.6%
May-2026	2026-05	S07	M. Sultana	$187,569	$233,626	16	+$46,057	Yes	124.6%
May-2026	2026-05	S08	J. Kabir	$314,205	$266,331	13	-$47,874	No	84.8%
Jun-2026	2026-06	S01	A. Rahman	$223,133	$266,804	14	+$43,671	Yes	119.6%
Jun-2026	2026-06	S02	N. Islam	$291,440	$309,366	18	+$17,926	Yes	106.2%
Jun-2026	2026-06	S03	S. Ahmed	$231,565	$184,777	11	-$46,788	No	79.8%
Jun-2026	2026-06	S04	T. Hossain	$330,440	$395,317	17	+$64,877	Yes	119.6%
Jun-2026	2026-06	S05	R. Karim	$320,078	$389,083	21	+$69,005	Yes	121.6%
Jun-2026	2026-06	S06	F. Hasan	$272,440	$294,612	16	+$22,172	Yes	108.1%
Jun-2026	2026-06	S07	M. Sultana	$378,091	$189,991	12	-$188,100	No	50.3%
Jun-2026	2026-06	S08	J. Kabir	$325,435	$513,211	21	+$187,776	Yes	157.7%
Jul-2026	2026-07	S01	A. Rahman	$409,660	$182,857	10	-$226,803	No	44.6%
Jul-2026	2026-07	S02	N. Islam	$329,517	$325,522	13	-$3,995	No	98.8%
Jul-2026	2026-07	S03	S. Ahmed	$207,306	$293,917	15	+$86,611	Yes	141.8%
Jul-2026	2026-07	S04	T. Hossain	$306,674	$260,620	13	-$46,054	No	85.0%

Task 2/3 - Target vs Actual by Salesperson and Month

Month	Month key	Salesperson_ID	Salesperson	Target (USD)	Actual (USD)	Orders	Variance (USD)	Met target?	Attainment %
Jul-2026	2026-07	S05	R. Karim	$185,117	$303,351	12	+$118,234	Yes	163.9%
Jul-2026	2026-07	S06	F. Hasan	$257,452	$332,191	16	+$74,739	Yes	129.0%
Jul-2026	2026-07	S07	M. Sultana	$263,526	$122,461	8	-$141,065	No	46.5%
Jul-2026	2026-07	S08	J. Kabir	$226,180	$295,804	11	+$69,624	Yes	130.8%
Aug-2026	2026-08	S01	A. Rahman	$279,862	$214,499	10	-$65,363	No	76.6%
Aug-2026	2026-08	S02	N. Islam	$396,859	$355,159	16	-$41,701	No	89.5%
Aug-2026	2026-08	S03	S. Ahmed	$280,614	$321,647	17	+$41,033	Yes	114.6%
Aug-2026	2026-08	S04	T. Hossain	$371,360	$345,693	16	-$25,667	No	93.1%
Aug-2026	2026-08	S05	R. Karim	$332,959	$198,032	14	-$134,927	No	59.5%
Aug-2026	2026-08	S06	F. Hasan	$197,518	$220,866	13	+$23,348	Yes	111.8%
Aug-2026	2026-08	S07	M. Sultana	$313,933	$211,908	11	-$102,025	No	67.5%
Aug-2026	2026-08	S08	J. Kabir	$391,481	$376,663	18	-$14,818	No	96.2%
Sep-2026	2026-09	S01	A. Rahman	$358,141	$157,039	8	-$201,102	No	43.8%
Sep-2026	2026-09	S02	N. Islam	$314,554	$320,916	15	+$6,362	Yes	102.0%
Sep-2026	2026-09	S03	S. Ahmed	$231,017	$411,168	17	+$180,151	Yes	178.0%
Sep-2026	2026-09	S04	T. Hossain	$233,633	$139,013	8	-$94,620	No	59.5%
Sep-2026	2026-09	S05	R. Karim	$190,694	$376,569	14	+$185,875	Yes	197.5%
Sep-2026	2026-09	S06	F. Hasan	$417,838	$263,751	15	-$154,087	No	63.1%
Sep-2026	2026-09	S07	M. Sultana	$242,801	$246,868	13	+$4,067	Yes	101.7%
Sep-2026	2026-09	S08	J. Kabir	$248,498	$326,341	17	+$77,843	Yes	131.3%
Total				**$21,110,246**	**$20,124,550**	**995**	**-$985,696**	**37 of 72**	**95.3%**

Attainment heatmap: salesperson x month (actual / target)

SP_ID	Salesperson	Jan-26	Feb-26	Mar-26	Apr-26	May-26	Jun-26	Jul-26	Aug-26	Sep-26	Jan-Sep

Task 2/3 - Target vs Actual by Salesperson and Month

Month	Month key	Salesperson_ID	Salesperson	Target (USD)	Actual (USD)	Orders	Variance (USD)	Met target?	Attainment %		
S01	A. Rahman	102.6%	129.2%	79.1%	13.3%	187.2%	119.6%	44.6%	76.6%	43.8%	**79.3%**
S02	N. Islam	108.2%	131.3%	146.8%	88.4%	100.0%	106.2%	98.8%	89.5%	102.0%	**106.9%**
S03	S. Ahmed	60.0%	38.3%	106.0%	99.7%	36.9%	79.8%	141.8%	114.6%	178.0%	**88.3%**
S04	T. Hossain	167.4%	93.8%	105.8%	101.0%	58.4%	119.6%	85.0%	93.1%	59.5%	**95.6%**
S05	R. Karim	51.4%	73.1%	112.3%	141.9%	36.2%	121.6%	163.9%	59.5%	197.5%	**95.2%**
S06	F. Hasan	160.7%	119.9%	93.9%	83.6%	32.6%	108.1%	129.0%	111.8%	63.1%	**92.1%**
S07	M. Sultana	135.3%	81.4%	214.5%	45.1%	124.6%	50.3%	46.5%	67.5%	101.7%	**87.9%**
S08	J. Kabir	126.3%	153.7%	127.9%	84.0%	84.8%	157.7%	130.8%	96.2%	131.3%	**118.1%**
All	**Team**	**110.6%**	**97.2%**	**116.7%**	**77.3%**	**70.4%**	**107.2%**	**96.9%**	**87.5%**	**100.2%**	**95.3%**

Red = below target, grey = on target, blue = above target. Targets exist only by salesperson x month, so they cannot be split by factory, product or customer region.

Task 4 - Advanced Analysis: What-If, Scenarios, Goal Seek, Sensitivity & Data Model

A. Baseline from the cleaned data (Jan-Sep 2026)

Baseline item	Value
Revenue (USD)	$20,124,550
Cost of goods (USD)	$8,807,842
Gross margin (USD)	$11,316,708
Gross margin %	56.2%
Units ordered	2,590,605
Average unit cost (USD)	$3.40
Defective units	64,756
Cost of defective units (USD)	$220,165
Revenue target Jan-Sep (USD)	$21,110,246
Target margin % (revenue-weighted)	57.4%

B. Scenario levers (edit the yellow cells - this replaces Scenario Manager)

Lever	Base	Conservative	Recovery plan	Stretch	Downside	Custom	Selected
Price change %	0.0%	+1.0%	+3.0%	+5.0%	-2.0%	0.0%	+3.0%
Volume change %	0.0%	-2.0%	+2.0%	+5.0%	-5.0%	0.0%	+2.0%
Unit cost change %	0.0%	-0.5%	-1.0%	-2.0%	+2.0%	0.0%	-1.0%
Defect reduction % (of defective units)	0.0%	+10.0%	+25.0%	+50.0%	0.0%	0.0%	+25.0%

Selected scenario (drop-down) Recovery plan — *Pick a scenario; the "Selected" column and section C update. "Custom" is the column to use with Goal Seek.*

C. Model outputs by scenario

Revenue = Rev x (1+price) x (1+volume); COGS = COGS x (1+volume) x (1+cost) - defect cost x (1+volume) x (1+cost) x defect reduction; Margin = Revenue - COGS.

Output	Base	Conservative	Recovery plan	Stretch	Downside	Custom	Selected
Revenue (USD)	$20,124,550	$19,919,280	$21,142,852	$22,187,316	$18,735,956	$20,124,550	**$21,142,852**
Cost of goods (USD)	$8,807,842	$8,567,058	$8,838,578	$8,949,994	$8,534,799	$8,807,842	**$8,838,578**
Gross margin (USD)	$11,316,708	$11,352,222	$12,304,274	$13,237,322	$10,201,158	$11,316,708	**$12,304,274**
Gross margin %	56.2%	57.0%	58.2%	59.7%	54.4%	56.2%	**58.2%**
Change in margin vs base (USD)	$0	+$35,513	+$987,566	+$1,920,614	-$1,115,551	$0	**+$987,566**
Target attainment %	95.3%	94.4%	100.2%	105.1%	88.8%	95.3%	**100.2%**
Margin gap vs target (pp)	-1.13%	-0.37%	+0.83%	+2.30%	-2.92%	-1.13%	**+0.83%**

D. Goal Seek answers (closed-form check of what Goal Seek returns)

Question	Set cell	To value	By changing	Answer (live)	Check: model at answer	Interpretation
Q1. Price change needed for Jan-Sep revenue to hit target (volume as in Custom)	H31	$21,110,246	H20	**+4.9%**	$21,110,246	Closing the $0.99M target gap by price alone needs ~+4.9% - not realistic in one step; combine with volume recovery in Apr-May
Q2. Price change needed to reach the revenue-weighted target margin	H34	57.4%	H20	**+2.7%**	57.36%	About +2.7% on price restores target margin, if buyers accept it.
Q3. Unit-cost change needed to reach target margin at current prices	H34	57.4%	H22	**-2.6%**	57.36%	Alternatively a ~2.6% unit-cost reduction (fabric, CM, rework) restores target margin without a price rise.

How to reproduce in Excel: Data > What-If Analysis > Goal Seek. Set cell / To value / By changing as listed (the Custom column). Goal Seek writes the answer into the Custom lever; it should equal the live answer column.

E. Sensitivity tables (two-variable Data Table equivalents)

Target attainment % (rows: price change, columns: volume change)

Price \ Volume	-10.0%	-5.0%	0.0%	+5.0%	+10.0%
-4.0%	82.4%	86.9%	91.5%	96.1%	100.7%
-2.0%	84.1%	88.8%	93.4%	98.1%	102.8%
0.0%	85.8%	90.6%	95.3%	100.1%	104.9%
+2.0%	87.5%	92.4%	97.2%	102.1%	107.0%
+4.0%	89.2%	94.2%	99.1%	104.1%	109.1%
+6.0%	90.9%	96.0%	101.1%	106.1%	111.2%

Gross margin % (rows: price change, columns: unit-cost change)

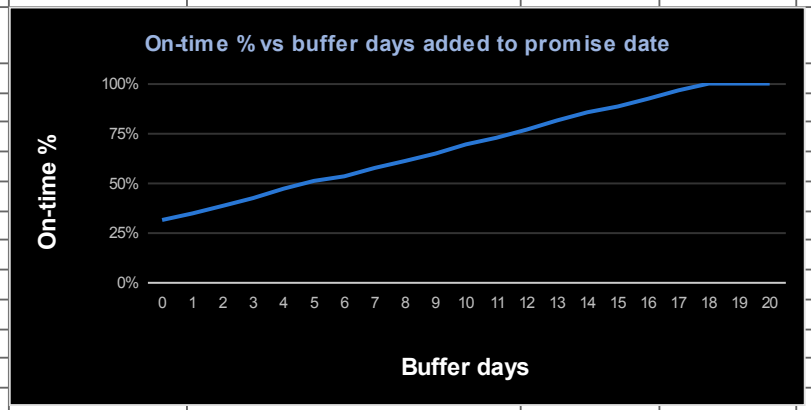
Price \ Unit cost	-4.0%	-2.0%	0.0%	+2.0%	+4.0%
-4.0%	56.2%	55.3%	54.4%	53.5%	52.6%
-2.0%	57.1%	56.2%	55.3%	54.4%	53.6%
0.0%	58.0%	57.1%	56.2%	55.4%	54.5%
+2.0%	58.8%	57.9%	57.1%	56.2%	55.4%
+4.0%	59.6%	58.8%	57.9%	57.1%	56.2%
+6.0%	60.4%	59.5%	58.7%	57.9%	57.1%

Equivalent to Data > What-If Analysis > Data Table with row input = volume/cost lever and column input = price lever. Baseline sits at 0% / 0%.

F. Delivery what-if: how many buffer days on the promise date would lift on-time delivery?

Target on-time %	**80.0%**	
Minimum buffer days needed	**13**	*Monotonic table, so the first buffer reaching the target = COUNTIF(on-time < target) + 1.*
Equivalent promise window (days)	58.1	*Either quote this window to buyers, or cut real lead time by the same number of days.*
On-time % achieved at that buffer	81.3%	

Buffer days added to promise	On-time %	Orders on time	Uplift vs today (pp)
0	31.2%	310	0.00%
1	34.5%	343	+3.32%
2	38.3%	381	+7.14%
3	42.2%	420	+11.06%
4	46.9%	467	+15.78%
5	50.9%	506	+19.70%
6	53.2%	529	+22.01%
7	57.4%	571	+26.23%
8	60.9%	606	+29.75%
9	64.6%	643	+33.47%
10	69.2%	689	+38.09%
11	72.6%	722	+41.41%
12	76.7%	763	+45.53%
13	81.3%	809	+50.15%
14	85.4%	850	+54.27%
15	88.3%	879	+57.19%
16	92.3%	918	+61.11%
17	96.5%	960	+65.33%
18	99.9%	994	+68.74%
19	99.9%	994	+68.74%
20	99.9%	994	+68.74%



G. Data model - relationships between datasets (Power Pivot / Data > Relationships)

From (many side)	To (one side)	Cardinality	Used for	Orphan rows (must be 0)
Clean_Orders[Customer_ID]	Customer_Master[Customer_ID]	Many-to-one	Region, customer type	0
Clean_Orders[Product_ID]	Product_Master[Product_ID]	Many-to-one	Standard price, target margin	0
Clean_Orders[Factory_ID]	Factory_Master[Factory_ID]	Many-to-one	Factory type, factory scorecard	0
Clean_Orders[Salesperson_ID]	Salesperson_Master[Salesperson_ID	Many-to-one	Salesperson views	0
Clean_Orders[Salesperson_ID & Order_Month]	Monthly_Targets[Salesperson_ID & M	Many-to-one (composite key)	Target vs actual (Target_Analysis)	0
Production_Data[Factory_ID]	Factory_Master[Factory_ID]	Many-to-one	Production KPIs in factory scorecard	0
Production_Data[Product_ID]	Product_Master[Product_ID]	Many-to-one	Production defect % by product	0

Excel route: format each sheet as a Table, Insert > PivotTable > "Add this data to the Data Model", then Data > Relationships. The composite target key needs a helper column (Salesperson_ID & "|" & Month) in both tables.

Dashboard engine - drives the Dashboard sheet (formulas only, no manual inputs)

Month list	Factory list	Category list	Region list	Salesperson list	Month start
All	All	All	All	All	
Jan 2026	Dhaka Knit Unit	Knit	Europe	A. Rahman	Jan-2026
Feb 2026	Gazipur Knit Unit	Woven	North America	N. Islam	Feb-2026
Mar 2026	Narayanganj Woven Unit		Asia	S. Ahmed	Mar-2026
Apr 2026	Chattogram Finishing Unit		Middle East	T. Hossain	Apr-2026
May 2026			Oceania	R. Karim	May-2026
Jun 2026				F. Hasan	Jun-2026
Jul 2026				M. Sultana	Jul-2026
Aug 2026				J. Kabir	Aug-2026
Sep 2026					Sep-2026

Filter	Selected	Criterion / from	to (month only)
Month	All	2000-01-01	2100-12-31
Factory	All	*	
Category	All	*	
Region	All	*	
Salesperson	All	*	
Targets comparable?	TRUE	*Targets exist only by salesperson x month*	

KPI (filtered)	Value
Revenue	$20,124,550
Gross margin	$11,316,708
Gross margin %	56.2%
Target margin %	57.4%
Orders	995

Units	2,590,605
On-time %	31.2%
Late orders	685
Late revenue	$13,875,460
Defect %	2.50%
Defective units	64,756
Rejected %	7.7%
Rejected orders	77
Rejected revenue	$1,583,938
Avg days late (late orders)	9.6
Fill rate	97.2%
Avg lead time (days)	50.6
Target (comparable only)	$21,110,246
Attainment %	95.3%
Share of Jan-Sep revenue	100.0%
Active filters	0

Month	Month start	Revenue	Target
Jan	Jan-2026	$2,260,036	$2,043,778
Feb	Feb-2026	$2,360,607	$2,429,595
Mar	Mar-2026	$2,692,551	$2,307,022
Apr	Apr-2026	$1,871,269	$2,421,701
May	May-2026	$1,794,070	$2,548,334
Jun	Jun-2026	$2,543,161	$2,372,622
Jul	Jul-2026	$2,116,724	$2,185,432
Aug	Aug-2026	$2,244,467	$2,564,586
Sep	Sep-2026	$2,241,665	$2,237,176

Product	Revenue	Rank helper
Denim Jeans	$2,480,669	2,480,668.810064
Work Jacket	$2,394,298	2,394,297.790065
Chino Trouser	$2,116,932	2,116,931.730066
Hoodie	$2,100,395	2,100,395.120067
Cargo Trouser	$1,958,031	1,958,030.860068
Dress Shirt	$1,945,533	1,945,533.130069
Leggings	$1,556,478	1,556,478.280070
Sweatshirt	$1,374,774	1,374,773.640071
Sports Tee	$1,283,442	1,283,441.530072
Polo Shirt	$1,281,787	1,281,786.690073
Basic T-Shirt	$1,035,362	1,035,362.450074
Kids T-Shirt	$596,850	596,850.040075

Factory	Short label	On-time %	Unit defect %	Rejected orders %	Orders	On-time target
Dhaka Knit Unit	Dhaka F01	29.1%	2.45%	7.3%	247	80.0%
Gazipur Knit Unit	Gazipur F02	31.0%	2.53%	8.8%	239	80.0%
Narayanganj Woven Unit	Narayanganj F03	31.6%	2.43%	6.3%	272	80.0%
Chattogram Finishing Unit	Chattogram F04	32.9%	2.61%	8.9%	237	80.0%

Region	Revenue
Europe	$9,788,638
North America	$4,115,723
Asia	$2,303,178
Middle East	$2,044,164
Oceania	$1,872,848

Salesperson	Actual	Target
A. Rahman	$2,241,832	$2,828,027
N. Islam	$2,965,202	$2,773,189
S. Ahmed	$2,279,921	$2,582,183
T. Hossain	$2,680,082	$2,803,963
R. Karim	$2,431,723	$2,555,429
F. Hasan	$2,316,217	$2,514,639
M. Sultana	$2,205,875	$2,510,005
J. Kabir	$3,003,700	$2,542,811

Quality status	Orders	Share
Passed	761	76.5%
Rework	157	15.8%
Rejected	77	7.7%

Delivery bucket	Orders	Share
On time / early	310	31.2%
1-7 days late	261	26.2%
8-14 days late	279	28.0%
15+ days late	145	14.6%

Rank	Product	Revenue	Share of filtered
1	Denim Jeans	$2,480,669	12.3%
2	Work Jacket	$2,394,298	11.9%
3	Chino Trouser	$2,116,932	10.5%

	4	Hoodie	$2,100,395	10.4%			
	5	Cargo Trouser	$1,958,031	9.7%			

[1] Raw_Row = source row in Raw_Orders (traceability). Cleaning_Note lists every change made to this row; changed cells are shaded orange and carry a comment with the original value.

[2] Lookup to the master tables, stored as values because the data sheets are sent separately.
Formula used (row 2): =INDEX(Customer_Master!$D$2:$D$21,MATCH(C2,Customer_Master!$A$2:$A$21,0))

[3] Lookup to the master tables, stored as values because the data sheets are sent separately.
Formula used (row 2): =INDEX(Customer_Master!$E$2:$E$21,MATCH(C2,Customer_Master!$A$2:$A$21,0))

[4] Lookup to the master tables, stored as values because the data sheets are sent separately.
Formula used (row 2): =INDEX(Factory_Master!$D$2:$D$5,MATCH(L2,Factory_Master!$A$2:$A$5,0))

[5] Lookup to the master tables, stored as values because the data sheets are sent separately.
Formula used (row 2): =INDEX(Product_Master!$G$2:$G$13,MATCH(F2,Product_Master!$A$2:$A$13,0))

[6] Lookup to the master tables, stored as values because the data sheets are sent separately.
Formula used (row 2): =INDEX(Product_Master!$H$2:$H$13,MATCH(F2,Product_Master!$A$2:$A$13,0))

[7] RAW: (blank)
CLEAN: Urban Loom Retail
RULE: Fill from Customer_Master via Customer_ID (INDEX/MATCH or XLOOKUP)

[8] RAW: Direct buyer
CLEAN: Direct Buyer
RULE: Standardise case (PROPER/Find & Replace)

[9] RAW: 2026/02/14
CLEAN: 2026-08-06
RULE: Convert text to a real date; value reconstructed as Actual_Ship_Date - Lead_Time_Days (2026-10-06 - 61)

[10] RAW: (blank)
CLEAN: S03
RULE: Fill from Salesperson_Master using the Salesperson name on the row

[11] RAW: passed
CLEAN: Passed
RULE: Standardise case (PROPER/Find & Replace)

[12] RAW: (blank)
CLEAN: 8.77

RULE: Impute as Revenue_USD / Order_Qty (33,554.02 / 3,826)

[13] RAW: s-03
CLEAN: S05
RULE: Re-key from the Salesperson name (R. Karim = S05)

[14] RAW: (blank)
CLEAN: (blank)
RULE: Leave blank; exclude this order from defect-rate denominators

[15] RAW: -250
CLEAN: 4397
RULE: Recover as Revenue_USD / Unit_Price_USD (53,995.16 / 12.28)

[16] RAW: 0
CLEAN: 10.44
RULE: Impute as Revenue_USD / Order_Qty (8,811.36 / 844)

[17] RAW: Buying house
CLEAN: Buying House
RULE: Standardise case (PROPER/Find & Replace)

[18] RAW: (blank)
CLEAN: 2026-02-20
RULE: Impute as Order_Date + Lead_Time_Days (2026-01-02 + 49)

[19] RAW: 15-06-2026
CLEAN: 2026-06-15
RULE: Convert dd-mm-yyyy text to a real date (DATE(RIGHT,MID,LEFT) or Text to Columns DMY)

[20] RAW: 9999999
CLEAN: 14613.7
RULE: Recalculate as Order_Qty x Unit_Price_USD (1,585 x 9.22)

[21] RAW: CUS999
CLEAN: CUS012
RULE: Re-key from Customer_Name via Customer_Master

[22] RAW: (blank)

CLEAN: 69
RULE: Recalculate as Actual_Ship_Date - Order_Date

[23] RAW: REWORK
CLEAN: Rework
RULE: Standardise case (PROPER/Find & Replace)

[24] RAW: PR999
CLEAN: PR008
RULE: Re-key from Product_Name via Product_Master

[25] RAW: 0
CLEAN: (blank)
RULE: Treat as unknown (blank); exclude from fill-rate denominators

[26] RAW: 2026-08-31
CLEAN: 2026-04-12
RULE: Convert text to a real date; value reconstructed as Order_Date + Lead_Time_Days (2026-02-08 + 63)

[27] RAW: (blank)
CLEAN: Unknown
RULE: Label as 'Unknown' (keep the row)

[28] RAW: F99
CLEAN: F04
RULE: Re-key from Factory name via Factory_Master

[29] RAW: 999
CLEAN: 65
RULE: Recalculate as Actual_Ship_Date - Order_Date

[30] RAW: S99
CLEAN: S04
RULE: Re-key from the Salesperson name via Salesperson_Master